

An Integrated Topological and Information-Geometric Generative Diffusion Framework for Robust Portfolio Optimization

A Continuous-Time Framework Based on Martingale
Schrödinger Bridges, Sliced Gromov-Wasserstein
Geometry, and Deep Backward SDE Control

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Abstract

Continuous-time portfolio optimization in volatile and non-stationary financial markets presents significant analytical and computational challenges at the intersection of quantitative finance, stochastic analysis, and statistical learning. Contemporary deep generative architectures—primarily Euclidean Denoising Diffusion Probabilistic Models (DDPMs) and continuous-time Score-based Generative Models (SGMs)—exhibit well-documented structural failure modes when applied to financial time series: (i) phase and temporal distortion introduced by discrete grid approximations or heuristic alignment algorithms, (ii) mathematical singularity and numerical instability of empirical covariance estimators during acute market distress ($\det(\Sigma) \rightarrow 0$), (iii) the synthesis of trajectories exhibiting persistent spurious drift, which violates the no-arbitrage condition of the Fundamental Theorem of Asset Pricing, and (iv) gradient conflict and computational intractability when coupling unsupervised score matching with downstream friction-aware stochastic optimal control under the curse of dimensionality.

This proposal sets forth an end-to-end continuous-time framework formulated across five interdependent stages. In **Phase 0**, an evidential continuous-time router governed by Neural Controlled Differential Equations (Neural CDEs) and Dirichlet priors continuously monitors incoming order flow, dynamically quantifying epistemic uncertainty via subjective logic to gate downstream representations without look-ahead bias. In **Phase 1**, temporal dynamics are mapped to coordinate-free topological invariants via persistent homology $(\beta_0, \beta_1, \beta_2)$ across Vietoris-Rips filtrations, yielding vectorized persistence landscapes residing in a reproducible Hilbert space. In **Phase 2**, cross-asset covariance structures are mapped to trace-normalized density operators on the Bures-Wasserstein Riemannian manifold; this metric avoids the boundary singularities of classical affine-invariant metrics and permits regularized tangent space projections. In the **Interface**, a Sliced Gromov-Wasserstein (SGW) optimal transport operator aligns the heterogeneous topological and covariance metric spaces with $\mathcal{O}(N \log N)$ complexity, synthesizing a unified conditioning tensor c_{unified} . In **Phase 3**, a jump-diffusion score-matching model augmented with compound Poisson jumps simulates heavy-tailed price dislocations, subject to an entropic Martingale Schrödinger Bridge (MSB) that mathematically eliminates arbitrage drift. Finally, in **Phase 4**, the friction-aware Hamilton-Jacobi-Bellman (HJB) system is mapped via the non-linear Feynman-Kac theorem into a Forward-Backward Stochastic Differential Equation (FBSDE), solved by a Deep BSDE controller under Huber-smoothed turnover transaction costs. The full system is stabilized through Two-Time-Scale Stochastic Approximation (TTSA), ensuring asymptotic convergence to a Stackelberg equilibrium between the generative score network and the allocation policy.

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Chapter 1

Mathematical Foundations and Theoretical Preliminaries

This chapter details the mathematical framework underpinning the proposed continuous-time generative portfolio architecture. The presentation develops systematically from continuous-time calculus and measure-theoretic probability through differential geometry, persistent homology, and Riemannian matrix manifolds, culminating in forward-backward stochastic differential equations and two-time-scale dynamical systems. Complete derivations and proofs are provided for all core foundational anchors.

1.1 Continuous Calculus, Measure-Theoretic Probability, and Spectral Analysis

1.1.1 Infinitesimal Calculus and Continuous Path Formulations

Financial asset dynamics in continuous-time trading environments cannot be modeled by discrete difference equations without introducing discretization biases and grid artifacts. We let $(\mathbb{R}, |\cdot|)$ denote the one-dimensional Euclidean space.

Definition 1.1 (Differentiability and Tangency). Let $f : (a, b) \rightarrow \mathbb{R}$ be a real-valued function. The derivative of f at $x \in (a, b)$, denoted $f'(x)$ or $\frac{df}{dx}$, is defined as the limit:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \quad (1.1)$$

provided this limit exists. In market microstructure analysis, when x denotes continuous trading time and $f(x)$ denotes cumulative signed order volume, $f'(x)$ corresponds to the instantaneous institutional order flow momentum.

Definition 1.2 (The Riemann-Stieltjes Integral). Let $f, \alpha : [a, b] \rightarrow \mathbb{R}$ be bounded functions. For a partition $\mathcal{P} = \{a = x_0 < x_1 < \cdots < x_n = b\}$ with mesh size $\|\mathcal{P}\| = \max_i(x_i - x_{i-1})$ and intermediate points $t_i \in [x_{i-1}, x_i]$, the Riemann-Stieltjes sum is:

$$S(\mathcal{P}, f, \alpha) = \sum_{i=1}^n f(t_i) [\alpha(x_i) - \alpha(x_{i-1})]. \quad (1.2)$$

If the limit as $\|\mathcal{P}\| \rightarrow 0$ exists independently of the partition and evaluation points, it is denoted $\int_a^b f(x) d\alpha(x)$. This formulation is essential when defining Neural Controlled

Differential Equations (Neural CDEs), where the integrator path $\mathbf{X}(t)$ is continuous and of bounded variation, but sampled from discrete event streams.

1.1.2 Measure Theory and Probability Foundations

Discrete combinatorial models are fundamentally insufficient for high-frequency limit order books and continuous stochastic price paths. A continuous model requires Kolmogorov's measure-theoretic formulation.

Definition 1.3 (Probability Space Triplet). A complete probability space is the triple $(\Omega, \mathcal{F}, \mathbb{P})$, where:

1. Ω is the **Sample Space**, the non-empty set of all elementary outcomes of the financial market universe (all realizations of price paths, order arrivals, and macro announcements).
2. $\mathcal{F}$ is a **σ -Algebra** on Ω : a non-empty family of subsets of Ω satisfying:

- $\Omega \in \mathcal{F}$,
- Closure under complementation: if $A \in \mathcal{F}$, then $A^c = \Omega \setminus A \in \mathcal{F}$,
- Closure under countable unions: if $\{A_i\}_{i=1}^{\infty} \subset \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

Elements of $\mathcal{F}$ are measurable events.

3. $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a **Probability Measure** satisfying Kolmogorov's axioms:

- Unit mass: $\mathbb{P}(\Omega) = 1$,
- Countable additivity: for any countable sequence of pairwise disjoint events $\{A_i\}_{i=1}^{\infty} \subset \mathcal{F}$,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i). \quad (1.3)$$

Definition 1.4 (Borel σ -Algebra and Random Variables). Let $\mathcal{B}(\mathbb{R}^N)$ denote the Borel σ -algebra on $\mathbb{R}^N$, defined as the smallest σ -algebra containing all open sets in $\mathbb{R}^N$. A multivariate **Random Variable** is an $\mathcal{F}/\mathcal{B}(\mathbb{R}^N)$ -measurable mapping $\mathbf{X} : \Omega \rightarrow \mathbb{R}^N$ such that:

$$\mathbf{X}^{-1}(B) = \{\omega \in \Omega : \mathbf{X}(\omega) \in B\} \in \mathcal{F}, \quad \forall B \in \mathcal{B}(\mathbb{R}^N). \quad (1.4)$$

The **Expected Value** is formalized as the Lebesgue integral with respect to $\mathbb{P}$:

$$\mathbb{E}[\mathbf{X}] = \int_{\Omega} \mathbf{X}(\omega) d\mathbb{P}(\omega). \quad (1.5)$$

1.1.3 Spectral Theory of Symmetric Positive Definite Matrices

Asset return covariances define the spatial geometry of portfolio theory. Understanding their geometric properties requires spectral matrix analysis.

Definition 1.5 (Symmetric Positive Definite Matrices). Let $\mathcal{S}^N$ denote the linear space of real symmetric $N \times N$ matrices ($A = A^\top$). A matrix $A \in \mathcal{S}^N$ is **Positive Semi-Definite** (PSD), denoted $A \succeq 0$, if $x^\top A x \geq 0$ for all $x \in \mathbb{R}^N$. It is **Symmetric Positive Definite** (SPD), denoted $A \succ 0$ (or $A \in \mathcal{S}_{++}^N$), if:

$$x^\top A x > 0, \quad \forall x \in \mathbb{R}^N \setminus \{0\}. \quad (1.6)$$

Theorem 1.1 (The Spectral Theorem for Real Symmetric Matrices). *Let $A \in \mathcal{S}^N$ be a real symmetric matrix. Then there exists an orthogonal matrix $Q \in \mathbb{R}^{N \times N}$ ($Q^\top Q = Q Q^\top = I_N$) and a real diagonal matrix $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_N)$ such that:*

$$A = Q \Lambda Q^\top = \sum_{i=1}^N \lambda_i q_i q_i^\top, \quad (1.7)$$

where $\{q_i\}_{i=1}^N$ are the orthonormal eigenvectors of A , and $\{\lambda_i\}_{i=1}^N$ are its eigenvalues. Furthermore, $A \succ 0$ if and only if $\lambda_i > 0$ for all $i \in \{1, \dots, N\}$.

Proof. We proceed by mathematical induction on the dimension N .

Base Step ($N = 1$): For a 1×1 matrix $A = [a_{11}]$, setting $Q = [1]$ and $\lambda_1 = a_{11} \in \mathbb{R}$ satisfies the condition trivially.

Inductive Hypothesis: Assume every symmetric matrix in $\mathcal{S}^{N-1}$ is orthogonally diagonalizable with real eigenvalues.

Inductive Step: Let $A \in \mathcal{S}^N$. By the Fundamental Theorem of Algebra, the characteristic polynomial $\det(\lambda I - A) = 0$ has at least one complex root $\lambda_1 \in \mathbb{C}$ with a corresponding non-zero complex eigenvector $v \in \mathbb{C}^N$ such that $Av = \lambda_1 v$. Taking the conjugate transpose:

$$v^* A = v^* A^\top = (Av)^* = (\lambda_1 v)^* = \bar{\lambda}_1 v^*. \quad (1.8)$$

Pre-multiplying $Av = \lambda_1 v$ by v^* yields $v^* Av = \lambda_1 v^* v = \lambda_1 \|v\|^2$. Post-multiplying $v^* A = \bar{\lambda}_1 v^*$ by v yields $v^* Av = \bar{\lambda}_1 v^* v = \bar{\lambda}_1 \|v\|^2$. Subtracting these equations gives:

$$(\lambda_1 - \bar{\lambda}_1) \|v\|^2 = 0. \quad (1.9)$$

Since $v \neq 0$, $\|v\|^2 > 0$, implying $\lambda_1 = \bar{\lambda}_1$, hence $\lambda_1 \in \mathbb{R}$. Consequently, there exists a normalized real eigenvector $q_1 \in \mathbb{R}^N$ such that $Aq_1 = \lambda_1 q_1$ with $\|q_1\|_2 = 1$.

By the Gram-Schmidt process, extend $\{q_1\}$ to an orthonormal basis $\{q_1, u_2, \dots, u_N\}$ for $\mathbb{R}^N$. Form the orthogonal matrix $P_1 = [q_1, U_1] \in \mathbb{R}^{N \times N}$. Then:

$$P_1^\top A P_1 = \begin{bmatrix} q_1^\top \\ U_1^\top \end{bmatrix} A \begin{bmatrix} q_1 & U_1 \end{bmatrix} = \begin{bmatrix} q_1^\top A q_1 & q_1^\top A U_1 \\ U_1^\top A q_1 & U_1^\top A U_1 \end{bmatrix}. \quad (1.10)$$

Since $Aq_1 = \lambda_1 q_1$, we have $q_1^\top A q_1 = \lambda_1 q_1^\top q_1 = \lambda_1$. Furthermore, $U_1^\top A q_1 = \lambda_1 U_1^\top q_1 = 0$ by orthogonality, and by symmetry $q_1^\top A U_1 = 0^\top$. Hence:

$$P_1^\top A P_1 = \begin{pmatrix} \lambda_1 & 0^\top \\ 0 & A_{N-1} \end{pmatrix}, \quad (1.11)$$

where $A_{N-1} = U_1^\top A U_1 \in \mathcal{S}^{N-1}$. By the induction hypothesis, there exists an orthogonal matrix $Q_{N-1} \in \mathbb{R}^{(N-1) \times (N-1)}$ such that $Q_{N-1}^\top A_{N-1} Q_{N-1} = \Lambda_{N-1}$ is diagonal. Defining:

$$Q = P_1 \begin{pmatrix} 1 & 0^\top \\ 0 & Q_{N-1} \end{pmatrix}, \quad (1.12)$$

we find $Q^\top Q = I_N$ and $Q^\top A Q = \text{diag}(\lambda_1, \Lambda_{N-1}) = \Lambda$. This completes the proof. $\square$

1.2 Filtrations, Conditional Expectations, and Riemannian Manifolds

1.2.1 Filtrations, Information Flow, and Martingales

In financial markets, information accumulates non-reversibly over time. To model adapted decision-making without look-ahead bias, we formalize filtrations.

Definition 1.6 (Filtration and Adapted Stochastic Processes). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space. A **Filtration** $\mathbb{F} = \{\mathcal{F}_t\}_{t \geq 0}$ is an increasing family of sub- σ -algebras of $\mathcal{F}$, such that:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}, \quad \forall 0 \leq s \leq t < \infty. \quad (1.13)$$

The sub- σ -algebra $\mathcal{F}_t$ represents the history of market events up to time t . A stochastic process $X = \{X_t\}_{t \geq 0}$ is **adapted** to $\mathbb{F}$ if X_t is $\mathcal{F}_t$ -measurable for each $t \geq 0$.

Definition 1.7 (Conditional Expectation via Radon-Nikodym). Let $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and let $\mathcal{G} \subset \mathcal{F}$ be a sub- σ -algebra. The **Conditional Expectation** of X given $\mathcal{G}$, denoted $\mathbb{E}[X \mid \mathcal{G}]$, is defined as the unique $\mathcal{G}$ -measurable random variable Y satisfying:

$$\int_G Y(\omega) d\mathbb{P}(\omega) = \int_G X(\omega) d\mathbb{P}(\omega), \quad \forall G \in \mathcal{G}. \quad (1.14)$$

Existence and uniqueness (almost surely) follow from the Radon-Nikodym theorem applied to the signed measure $\nu(G) = \int_G X d\mathbb{P}$ with respect to $\mathbb{P}|_{\mathcal{G}}$. Geometrically, for $X \in L^2$, $\mathbb{E}[X \mid \mathcal{G}]$ is the orthogonal projection of X onto $L^2(\Omega, \mathcal{G}, \mathbb{P})$.

Definition 1.8 (Martingales). An adapted stochastic process $M = \{M_t\}_{t \geq 0}$ with $\mathbb{E}[|M_t|] < \infty$ is a **Martingale** with respect to $(\mathbb{F}, \mathbb{P})$ if:

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s, \quad \forall 0 \leq s \leq t. \quad (1.15)$$

Under the Fundamental Theorem of Asset Pricing, the absence of arbitrage is equivalent to the existence of an equivalent probability measure under which discounted price processes are local martingales.

1.2.2 Metric Spaces and Riemannian Manifolds

Standard neural architectures assume data reside in flat Euclidean space $\mathbb{R}^D$ with canonical inner product $\langle u, v \rangle = u^\top v$. However, financial covariance matrices and normalized density operators reside on curved Riemannian manifolds.

Definition 1.9 (Metric Space). A metric space is a pair (M, d) , where M is a set and $d : M \times M \rightarrow [0, \infty)$ is a metric satisfying identity of indiscernibles ($d(x, y) = 0 \iff x = y$), symmetry ($d(x, y) = d(y, x)$), and the triangle inequality ($d(x, z) \leq d(x, y) + d(y, z)$).

Definition 1.10 (Riemannian Manifold and Tangent Space). A smooth manifold $\mathcal{M}$ of dimension d is a topological Hausdorff space locally homeomorphic to $\mathbb{R}^d$. At each point $p \in \mathcal{M}$, the **Tangent Space** $T_p \mathcal{M}$ is a d -dimensional vector space containing the velocity vectors of all smooth curves through p . A **Riemannian Metric** g is a smoothly varying family of inner products $g_p : T_p \mathcal{M} \times T_p \mathcal{M} \rightarrow \mathbb{R}$. The Riemannian distance between

$p, q \in \mathcal{M}$ is the infimum of the lengths of all piecewise smooth curves $\gamma : [0, 1] \rightarrow \mathcal{M}$ connecting them:

$$d_{\mathcal{M}}(p, q) = \inf_{\gamma} \int_0^1 \sqrt{g_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t))} dt. \quad (1.16)$$

Curves realizing this infimum are geodesics.

1.3 Stochastic Calculus, Algebraic Topology, and Information Geometry

1.3.1 Brownian Motion, Quadratic Variation, and Itô's Lemma

Continuous-time market diffusion is driven microscopically by standard Brownian motion.

Definition 1.11 (Standard Brownian Motion). A continuous stochastic process $W = \{W_t\}_{t \geq 0}$ is a standard 1-dimensional Brownian motion (Wiener process) if $W_0 = 0$ a.s., increments are independent, $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $0 \leq s < t$, and sample paths $t \mapsto W_t$ are almost surely continuous.

Theorem 1.2 (Quadratic Variation of Brownian Motion). Let $\Pi_n = \{0 = t_0 < t_1 < \dots < t_n = t\}$ be a sequence of partitions of $[0, t]$ with mesh $\|\Pi_n\| \rightarrow 0$ as $n \rightarrow \infty$. The quadratic variation of W over $[0, t]$ converges in $L^2(\Omega)$ to t :

$$[W]_t = \lim_{\|\Pi_n\| \rightarrow 0} \sum_{i=1}^n (W_{t_i} - W_{t_{i-1}})^2 = t. \quad (1.17)$$

This formally establishes the standard Itô rule: $(dW_t)^2 = dt$, while $(dt)^2 = 0$ and $dt dW_t = 0$.

Theorem 1.3 (Itô's Lemma for Drift-Diffusion Processes). Let X_t be an Itô process satisfying:

$$dX_t = \mu(X_t, t)dt + \sigma(X_t, t)dW_t. \quad (1.18)$$

Let $f(x, t) : \mathbb{R} \times [0, \infty) \rightarrow \mathbb{R}$ be $C^{2,1}$. Then $Y_t = f(X_t, t)$ is also an Itô process, and its differential is:

$$df(X_t, t) = \left[\frac{\partial f}{\partial t} + \mu(X_t, t) \frac{\partial f}{\partial x} + \frac{1}{2} \sigma(X_t, t)^2 \frac{\partial^2 f}{\partial x^2} \right] dt + \sigma(X_t, t) \frac{\partial f}{\partial x} dW_t. \quad (1.19)$$

Proof. Let $\Pi_n = \{0 = t_0 < t_1 < \dots < t_n = t\}$ be a partition of $[0, t]$. Expand the telescoping sum:

$$f(X_t, t) - f(X_0, 0) = \sum_{i=0}^{n-1} [f(X_{t_{i+1}}, t_{i+1}) - f(X_{t_i}, t_i)]. \quad (1.20)$$

Applying a second-order multivariate Taylor expansion for increments $\Delta t_i = t_{i+1} - t_i$ and $\Delta X_i = X_{t_{i+1}} - X_{t_i}$:

$$\begin{aligned} f(X_{t_{i+1}}, t_{i+1}) - f(X_{t_i}, t_i) &= \frac{\partial f}{\partial t} \Delta t_i + \frac{\partial f}{\partial x} \Delta X_i + \frac{1}{2} \frac{\partial^2 f}{\partial t^2} (\Delta t_i)^2 + \frac{\partial^2 f}{\partial t \partial x} \Delta t_i \Delta X_i \\ &\quad + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} (\Delta X_i)^2 + R_i. \end{aligned} \quad (1.21)$$

Substituting $\Delta X_i = \mu_i \Delta t_i + \sigma_i \Delta W_i$:

$$(\Delta X_i)^2 = \mu_i^2 (\Delta t_i)^2 + 2\mu_i \sigma_i \Delta t_i \Delta W_i + \sigma_i^2 (\Delta W_i)^2. \quad (1.22)$$

As $\|\Pi_n\| \rightarrow 0$:

1. Terms in $(\Delta t_i)^2$ and $\Delta t_i \Delta W_i$ vanish in L^2 since $\sum (\Delta t_i)^2 \leq t \max \Delta t_i \rightarrow 0$.
2. By Theorem 1.2, $\sum_{i=0}^{n-1} \sigma_i^2 (\Delta W_i)^2$ converges in probability to $\int_0^t \sigma_s^2 ds$.
3. The remainder terms satisfy $\sum R_i \rightarrow 0$.

Summing and passing to the limit yields:

$$f(X_t, t) - f(X_0, 0) = \int_0^t \left[\frac{\partial f}{\partial s} + \mu_s \frac{\partial f}{\partial x} + \frac{1}{2} \sigma_s^2 \frac{\partial^2 f}{\partial x^2} \right] ds + \int_0^t \sigma_s \frac{\partial f}{\partial x} dW_s, \quad (1.23)$$

which is the stated differential form. $\square$

1.3.2 Lévy Jump Processes and Discontinuous Market Dynamics

Pure diffusion models assume sample-path continuity. However, high-frequency order books experience discrete liquidity disruptions and jump events.

Definition 1.12 (Compound Poisson Process). A Poisson process $N = \{N_t\}_{t \geq 0}$ with intensity $\lambda > 0$ satisfies:

$$\mathbb{P}(N_t - N_s = k) = \frac{(\lambda(t-s))^k}{k!} e^{-\lambda(t-s)}, \quad k \in \mathbb{N}_0. \quad (1.24)$$

A **Compound Poisson Process** is $J_t = \sum_{i=1}^{N_t} Y_i$, where $\{Y_i\}_{i=1}^\infty$ are i.i.d. jump amplitudes independent of N_t . The compensated process $\tilde{N}_t = N_t - \lambda t$ is a martingale.

Theorem 1.4 (Lévy-Itô Decomposition). *Every Lévy process X_t with stationary, independent increments starting at zero admits the unique decomposition:*

$$X_t = bt + \sigma W_t + \int_{|x| < 1} x [N(t, dx) - t\nu(dx)] + \int_{|x| \geq 1} x N(t, dx), \quad (1.25)$$

where $b \in \mathbb{R}$, $\sigma \geq 0$, W_t is standard Brownian motion, $N(t, dx)$ is a Poisson random measure on $\mathbb{R} \times [0, \infty)$, and $\nu(dx)$ is the Lévy jump measure satisfying $\int_{\mathbb{R}} \min(1, x^2) \nu(dx) < \infty$.

1.3.3 Algebraic Topology and Persistent Homology

Linear transforms often fail to capture geometric invariants in complex financial phase spaces. Algebraic topology characterizes these structures via simplicial complexes.

Definition 1.13 (Simplex and Simplicial Complex). Let $V = \{v_0, v_1, \dots, v_k\} \subset \mathbb{R}^D$ be affinely independent. The k -**Simplex** $\sigma = [v_0, \dots, v_k]$ is:

$$\sigma = \left\{ \sum_{i=0}^k \alpha_i v_i : \sum_{i=0}^k \alpha_i = 1, \alpha_i \geq 0 \right\}. \quad (1.26)$$

A **Simplicial Complex** $\mathcal{K}$ is a finite collection of simplices such that any face of a simplex in $\mathcal{K}$ belongs to $\mathcal{K}$, and the non-empty intersection of any two simplices in $\mathcal{K}$ is a common face.

Definition 1.14 (Vietoris-Rips Complex and Filtration). Given a finite point cloud $P \subset \mathbb{R}^D$ and scale $\epsilon \geq 0$, the **Vietoris-Rips Complex** is:

$$\text{VR}(P, \epsilon) = \{\sigma \subseteq P : d(u, v) \leq \epsilon, \forall u, v \in \sigma\}. \quad (1.27)$$

A **Filtration** is a nested family parameterized by ϵ :

$$\emptyset = \text{VR}(P, 0) \subseteq \text{VR}(P, \epsilon_1) \subseteq \cdots \subseteq \text{VR}(P, \epsilon_{\max}) = \mathcal{K}. \quad (1.28)$$

Definition 1.15 (Boundary Operators and Homology Groups). Let $C_k(\mathcal{K})$ denote the vector space of k -chains over $\mathbb{Z}_2$. The **Boundary Operator** $\partial_k : C_k(\mathcal{K}) \rightarrow C_{k-1}(\mathcal{K})$ is defined on simplices by:

$$\partial_k[v_0, \dots, v_k] = \sum_{i=0}^k [v_0, \dots, \hat{v}_i, \dots, v_k], \quad (1.29)$$

where $\hat{v}_i$ denotes omission. It satisfies $\partial_k \circ \partial_{k+1} = 0$. Consequently, boundaries $B_k = \text{im}(\partial_{k+1})$ form a subspace of cycles $Z_k = \ker(\partial_k)$. The **k -th Homology Group** is:

$$H_k(\mathcal{K}) = Z_k / B_k = \ker(\partial_k) / \text{im}(\partial_{k+1}), \quad (1.30)$$

with dimension $\beta_k = \dim(H_k)$ giving the **k -th Betti Number** (β_0 : components, β_1 : 1D loops, β_2 : 2D cavities).

Definition 1.16 (Persistence Landscapes). Given a persistence diagram $\mathcal{D} = \{(b_i, d_i)\}_{i \in I}$, define the triangular function:

$$f_{(b_i, d_i)}(t) = \begin{cases} t - b_i & \text{if } t \in [b_i, \frac{b_i + d_i}{2}], \\ d_i - t & \text{if } t \in (\frac{b_i + d_i}{2}, d_i], \\ 0 & \text{otherwise.} \end{cases} \quad (1.31)$$

The **k -th Persistence Landscape** $\lambda_k(t)$ is the k -th largest value of $\{f_{(b_i, d_i)}(t)\}_{i \in I}$. Because $\lambda_k \in L^p(\mathbb{R})$, this transformation vectorizes topological invariants into a Hilbert space suitable for machine learning models.

1.3.4 Information Geometry on the PSD Cone and the Bures Metric

Standard portfolio optimization requires inverting the empirical covariance matrix Σ^{-1} . During systemic shocks, cross-asset correlations spike, driving Σ toward singularity ($\det(\Sigma) \rightarrow 0$) and causing numerical divergence in classical algorithms. Information geometry on trace-normalized matrices resolves this issue.

Definition 1.17 (Normalized Covariance States). Let $\Sigma_t \in \mathcal{S}_{++}^N$ be an empirical covariance matrix. The **Normalized Covariance State** ρ_t is:

$$\rho_t = \frac{\Sigma_t}{\text{Tr}(\Sigma_t)}. \quad (1.32)$$

By construction, ρ_t satisfies: (i) real symmetry: $\rho_t = \rho_t^\top$, (ii) positive semi-definiteness: $\rho_t \succeq 0$, and (iii) unit trace: $\text{Tr}(\rho_t) = 1$. The von Neumann entropy $S(\rho) = -\text{Tr}(\rho \log \rho)$ quantifies systemic dispersion across assets.

Definition 1.18 (Uhlmann Fidelity and Bures Metric). For density matrices $\rho_1, \rho_2 \in \mathcal{S}_{++}^N$, the **Fidelity** $\mathcal{F}(\rho_1, \rho_2)$ is:

$$\mathcal{F}(\rho_1, \rho_2) = \left(\text{Tr} \left[\sqrt{\sqrt{\rho_1} \rho_2 \sqrt{\rho_1}} \right] \right)^2 \in [0, 1]. \quad (1.33)$$

The **Bures Metric** $D_B(\rho_1, \rho_2)$ is the induced Riemannian distance:

$$D_B(\rho_1, \rho_2) = \sqrt{2 \left(1 - \sqrt{\mathcal{F}(\rho_1, \rho_2)} \right)} = \sqrt{2 \left(1 - \text{Tr} \sqrt{\sqrt{\rho_1} \rho_2 \sqrt{\rho_1}} \right)}. \quad (1.34)$$

Theorem 1.5 (Non-Singularity of the Bures Metric on the PSD Boundary). *Let $\{\Sigma_k\}_{k=1}^\infty \subset \mathcal{S}_{++}^N$ be a sequence of covariance matrices converging to a rank-deficient matrix Σ_* ($\text{rank}(\Sigma_*) = r < N$, so $\det(\Sigma_*) = 0$). While the classical Fisher information metric and precision matrix diverge ($\|\Sigma_k^{-1}\|_F \rightarrow \infty$), the Bures distance $D_B(\rho_k, \rho_*)$ remains bounded in $[0, \sqrt{2}]$, continuous, and differentiable on $\mathcal{S}_+^N$.*

Proof. Let $\rho_1, \rho_2 \in \mathcal{S}_+^N$ be any two density matrices, possibly singular. The principal matrix square root $A = \sqrt{\rho_1}$ is unique via the spectral theorem: $A = Q\Lambda^{1/2}Q^\top \succeq 0$. The product $M = A\rho_2A = \sqrt{\rho_1}\rho_2\sqrt{\rho_1}$ is symmetric positive semi-definite, since for any $x \in \mathbb{R}^N$:

$$x^\top Mx = x^\top A\rho_2Ax = (Ax)^\top \rho_2(Ax) \geq 0. \quad (1.35)$$

Hence, all eigenvalues μ_i of M are non-negative real numbers, and $\sqrt{M}$ has real eigenvalues $\sqrt{\mu_i} \geq 0$. The trace is $\text{Tr}(\sqrt{M}) = \sum_{i=1}^N \sqrt{\mu_i}$. By the Cauchy-Schwarz inequality for trace-class operators:

$$0 \leq \text{Tr} \sqrt{\sqrt{\rho_1} \rho_2 \sqrt{\rho_1}} \leq \sqrt{\text{Tr}(\rho_1) \text{Tr}(\rho_2)} = 1. \quad (1.36)$$

Substituting this into the definition of D_B :

$$0 \leq D_B(\rho_1, \rho_2) = \sqrt{2 \left(1 - \text{Tr} \sqrt{M} \right)} \leq \sqrt{2}. \quad (1.37)$$

Computing D_B requires only matrix multiplications and principal square roots; it involves no matrix inversions. Consequently, the distance remains bounded and differentiable on the boundary of singular matrices, avoiding the rank-collapse singularities of classical metrics during market crises. $\square$

1.4 Advanced Optimization, Optimal Transport, and Stochastic Control

1.4.1 Sliced Gromov-Wasserstein Optimal Transport

Aligning discrete persistence landscapes $\mathcal{T}_t$ with continuous covariance manifolds ρ_t cannot be accomplished via Euclidean concatenation. The Gromov-Wasserstein distance measures discrepancies across distinct metric spaces.

Definition 1.19 (Gromov-Wasserstein Distance). Let $(\mathcal{X}, d_X, \mu)$ and $(\mathcal{Y}, d_Y, \nu)$ be two metric measure spaces. The **Gromov-Wasserstein (GW)** distance is:

$$\text{GW}_2^2(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{Y}} \int_{\mathcal{X} \times \mathcal{Y}} |d_X(x, x') - d_Y(y, y')|^2 d\pi(x, y) d\pi(x', y'), \quad (1.38)$$

where $\Pi(\mu, \nu)$ is the set of joint couplings. Solving this directly is a Quadratic Assignment Problem with $\mathcal{O}(N^3)$ complexity, which is impractical for real-time allocation.

Definition 1.20 (Sliced Gromov-Wasserstein Formulation). Via the Radon transform, multi-dimensional point sets can be projected onto one-dimensional lines. For the unit sphere $\mathbb{S}^{d-1}$, the **Sliced Gromov-Wasserstein (SGW)** distance is:

$$\text{SGW}(\mu, \nu) = \int_{\mathbb{S}^{d-1}} \text{GW}_{1D}(\langle \theta, \mu \rangle, \langle \theta, \nu \rangle) d\sigma(\theta). \quad (1.39)$$

In one dimension, the optimal coupling π^* is obtained in closed form by sorting projected coordinates:

$$\text{GW}_{1D}(\mu_\theta, \nu_\theta) = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \left| |x_{\sigma(i)}^\theta - x_{\sigma(j)}^\theta| - |y_{\tau(i)}^\theta - y_{\tau(j)}^\theta| \right|^2, \quad (1.40)$$

where σ, τ sort the projected vectors. This reduces the computational complexity from $\mathcal{O}(N^3)$ to $\mathcal{O}(P \cdot N \log N)$, where P is the number of random slices.

1.4.2 Score-Based Diffusion and Martingale Schrödinger Bridges

Score-based generative models synthesize trajectories by reversing an Itô diffusion process.

Theorem 1.6 (Anderson's Reverse-Time SDE Theorem). *Let $Y_t \in \mathbb{R}^N$ satisfy the forward SDE:*

$$dY_t = f(Y_t, t)dt + g(t)dW_t, \quad t \in [0, T]. \quad (1.41)$$

Under standard regularity conditions on f and g , the time-reversed process $\bar{Y}_t = Y_{T-t}$ satisfies:

$$d\bar{Y}_t = [-f(\bar{Y}_t, T-t) + g(T-t)^2 \nabla_Y \log p_{T-t}(\bar{Y}_t)] dt + g(T-t)d\bar{W}_t, \quad (1.42)$$

where $\bar{W}_t$ is a backward Brownian motion, and $\nabla_Y \log p_t(Y)$ is the spatial Stein score function.

Definition 1.21 (Martingale Schrödinger Bridge). Let $\mathbb{P}_{\text{ref}}$ be a reference jump-diffusion on $\Omega = C([0, T], \mathbb{R}^N)$. The **Martingale Schrödinger Bridge (MSB)** problem is the constrained entropic optimal transport problem:

$$\inf_{\mathbb{P} \in \mathcal{P}(\Omega)} \text{KL}(\mathbb{P} \parallel \mathbb{P}_{\text{ref}}), \quad (1.43)$$

subject to marginal constraints $\mathbb{P}_0 = \mu_0$, $\mathbb{P}_T = \mu_T$, and the **Martingale Constraint**:

$$\mathbb{E}^\mathbb{P}[P_t \mid \mathcal{F}_s] = P_s e^{r_f(t-s)}, \quad \forall 0 \leq s \leq t \leq T, \quad (1.44)$$

where $P_t = \exp(Y_t)$ are asset prices. This constraint aligns the generative drift with the risk-neutral measure, eliminating uncompensated arbitrage drift in synthetic paths.

1.4.3 Deep Backward SDEs and the Non-Linear Feynman-Kac Theorem

Dynamic asset allocation under trading friction requires solving the Hamilton-Jacobi-Bellman (HJB) PDE. In high dimensions ($N = 500$), classical spatial discretizations suffer from the curse of dimensionality ($\mathcal{O}(K^{500})$). Deep Backward SDEs (Deep BSDE) circumvent this limitation.

Theorem 1.7 (Non-Linear Feynman-Kac Theorem). *Consider the semilinear parabolic PDE:*

$$\frac{\partial V}{\partial t}(t, x) + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t, x) \nabla_x^2 V(t, x)) + \nabla_x V(t, x)^\top \mu(t, x) + h(t, x, V(t, x), \sigma^\top(t, x) \nabla_x V(t, x)) = 0, \quad (1.45)$$

with terminal condition $V(T, x) = g(x)$. Let $(X_t)_{t \in [0, T]}$ be the forward process:

$$dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t. \quad (1.46)$$

Define the backward processes:

$$Y_t = V(t, X_t) \in \mathbb{R}, \quad Z_t = \sigma^\top(t, X_t) \nabla_x V(t, X_t) \in \mathbb{R}^d. \quad (1.47)$$

Then (X_t, Y_t, Z_t) uniquely solves the Forward-Backward Stochastic Differential Equation (FBSDE):

$$dY_t = -h(t, X_t, Y_t, Z_t)dt + Z_t^\top dW_t, \quad Y_T = g(X_T). \quad (1.48)$$

Proof. Apply Itô's Lemma (Theorem 1.3) to $V(t, X_t)$:

$$\begin{aligned} dV(t, X_t) &= \frac{\partial V}{\partial t}dt + \nabla_x V^\top dX_t + \frac{1}{2} \text{Tr}(dX_t dX_t^\top \nabla_x^2 V) \\ &= \left[\frac{\partial V}{\partial t} + \nabla_x V^\top \mu + \frac{1}{2} \text{Tr}(\sigma \sigma^\top \nabla_x^2 V) \right] dt + \nabla_x V^\top \sigma dW_t. \end{aligned} \quad (1.49)$$

From the semilinear PDE, the bracketed drift simplifies to:

$$\frac{\partial V}{\partial t} + \nabla_x V^\top \mu + \frac{1}{2} \text{Tr}(\sigma \sigma^\top \nabla_x^2 V) = -h(t, X_t, V(t, X_t), \sigma^\top \nabla_x V). \quad (1.50)$$

Setting $Y_t = V(t, X_t)$ and $Z_t = \sigma^\top \nabla_x V(t, X_t)$ yields:

$$dY_t = -h(t, X_t, Y_t, Z_t)dt + Z_t^\top dW_t, \quad (1.51)$$

with $Y_T = V(T, X_T) = g(X_T)$. This maps the high-dimensional PDE into a coupled forward-backward SDE that can be solved via Monte Carlo simulation and neural network approximations. $\square$

1.4.4 Two-Time-Scale Stochastic Approximation and Borkar's Theorem

When optimizing a system comprising both a generative network (θ) and a policy controller (ω), single-timescale gradient descent often leads to gradient interference ($\langle \nabla_\theta \mathcal{L}_{\text{gen}}, \nabla_\omega \mathcal{L}_{\text{policy}} \rangle < 0$). Two-Time-Scale Stochastic Approximation (TTSA) decouples their training dynamics.

Theorem 1.8 (Borkar's Two-Time-Scale Convergence Theorem). *Consider the coupled stochastic recursions:*

$$\theta_{k+1} = \theta_k + \alpha_k \left[f(\theta_k, \omega_k) + M_{k+1}^{(1)} \right], \quad (1.52)$$

$$\omega_{k+1} = \omega_k + \beta_k \left[g(\theta_k, \omega_k) + M_{k+1}^{(2)} \right], \quad (1.53)$$

where $M_{k+1}^{(1)}, M_{k+1}^{(2)}$ are martingale difference sequences. Assume:

1. *Step-size conditions:* $\sum_k \alpha_k = \infty$, $\sum_k \beta_k = \infty$, $\sum_k \alpha_k^2 < \infty$, $\sum_k \beta_k^2 < \infty$.
2. *Timescale separation:* $\lim_{k \rightarrow \infty} \frac{\beta_k}{\alpha_k} = 0$.
3. *For each fixed ω , the ODE $\dot{\theta}(t) = f(\theta(t), \omega)$ has a unique globally asymptotically stable equilibrium $\lambda(\omega)$, with λ Lipschitz continuous.*
4. *The slow ODE $\dot{\omega}(t) = g(\lambda(\omega(t)), \omega(t))$ has a globally asymptotically stable equilibrium ω^* .*

Then $(\theta_k, \omega_k) \rightarrow (\lambda(\omega^*), \omega^*)$ almost surely as $k \rightarrow \infty$.

Proof. Let $\tau_n = \sum_{i=0}^{n-1} \alpha_i$ define the continuous interpolated trajectory for the fast recursion (1.52). On the fast timescale α_k , the ratio $\beta_k/\alpha_k \rightarrow 0$. Dividing (1.53) by α_k :

$$\frac{\omega_{k+1} - \omega_k}{\alpha_k} = \frac{\beta_k}{\alpha_k} \left[g(\theta_k, \omega_k) + M_{k+1}^{(2)} \right] \rightarrow 0 \quad \text{a.s.} \quad (1.54)$$

Consequently, ω appears quasi-static on the fast timescale. By standard stochastic approximation theory, θ_k tracks the ODE $\dot{\theta}(t) = f(\theta(t), \omega)$, converging to $\lambda(\omega_k)$.

Now consider the slow recursion (1.53) on timescale $\sigma_n = \sum_{i=0}^{n-1} \beta_i$. Because $\|\theta_k - \lambda(\omega_k)\| \rightarrow 0$, the update can be written as:

$$\omega_{k+1} = \omega_k + \beta_k \left[g(\lambda(\omega_k), \omega_k) + \epsilon_k + M_{k+1}^{(2)} \right], \quad (1.55)$$

where $\epsilon_k = g(\theta_k, \omega_k) - g(\lambda(\omega_k), \omega_k) \rightarrow 0$ almost surely by the Lipschitz continuity of g . This perturbed system asymptotically tracks the slow ODE:

$$\dot{\omega}(t) = g(\lambda(\omega(t)), \omega(t)). \quad (1.56)$$

By Assumption 4, this ODE converges to ω^* . Thus $\omega_k \rightarrow \omega^*$ almost surely, and $\theta_k \rightarrow \lambda(\omega^*)$ almost surely. $\square$

Chapter 2

Introduction and Research Overview

2.1 Background: From Point Forecasting to Generative Modeling

For several decades, quantitative finance focused primarily on point forecasting and conditional mean estimation. Classical econometrics relied on autoregressive moving average (ARMA) models and GARCH volatility frameworks. With the advent of deep learning, these were expanded to Recurrent Neural Networks (RNNs), LSTMs, and attention-based Transformer models.

Despite increasing capacity, these models generally formulate prediction as conditional mean regression:

$$\hat{Y}_{t+1} = \operatorname{argmin}_{\hat{Y}} \mathbb{E} \left[\|Y_{t+1} - \hat{Y}\|_2^2 \mid \mathcal{F}_t \right]. \quad (2.1)$$

This formulation assumes that financial returns can be adequately represented by a conditional expectation with additive symmetric noise. In reality, electronic order books and multi-asset markets exhibit non-equilibrium dynamics: reflexivity, feedback loops, liquidity dry-ups, and asynchronous trading. Under these conditions, the future return distribution $p(Y_{t+1} \mid \mathcal{F}_t)$ is non-Gaussian, multi-modal, and heavy-tailed. A single point forecast averages out critical tail realizations, which can lead to severe portfolio misallocations during periods of market stress.

Generative modeling—specifically Denoising Diffusion Probabilistic Models (DDPMs) and Score-based Generative Models (SGMs)—offers an alternative paradigm. Rather than predicting a single mean path, diffusion models learn the conditional distribution $p(Y_{t+1} \mid \mathcal{F}_t)$ by reversing a continuous-time stochastic process that perturbs data into Gaussian noise. In portfolio optimization, this permits the simulation of consistent market scenarios that capture cross-asset dependencies and tail co-movements.

2.2 Motivation: Structural Challenges in Financial Diffusion

Applying computer-vision diffusion models directly to financial markets introduces significant methodological limitations:

1. **Low Signal-to-Noise Ratio and Microstructure Noise:** Unlike natural images with clear spatial hierarchies, high-frequency financial time series are dominated

by microstructure noise. Standard diffusion models tend to fit this high-frequency noise rather than extracting macroeconomic drift.

2. **Covariance Singularity during Distress:** During systemic crises, cross-asset correlations rise sharply, rendering the empirical covariance matrix Σ ill-conditioned ($\det(\Sigma) \rightarrow 0$). Standard diffusion models and mean-variance optimizers that depend on Σ^{-1} experience numerical instability.
3. **Arbitrage Hallucinations:** Unconstrained diffusion models optimize likelihood bounds without imposing economic constraints. Consequently, they often generate synthetic paths with persistent non-martingale drift in the absence of risk premiums, implying unrealistic arbitrage opportunities.
4. **Computational Bottlenecks in Stochastic Control:** Dynamic portfolio control under transaction costs requires solving high-dimensional HJB equations, which scale exponentially with the number of assets.

To address these challenges, this proposal introduces:

- **Topological Data Analysis (TDA)** to extract deformation-invariant temporal features without discrete alignment artifacts,
- **Information Geometry via the Bures Metric** to ensure computational stability on singular covariance boundaries,
- **Martingale Schrödinger Bridges (MSB)** to enforce no-arbitrage constraints on generative drift,
- **Deep Backward SDEs (Deep BSDE)** with **Two-Time-Scale Stochastic Approximation (TTSA)** to solve high-dimensional friction-aware portfolio control.

2.3 Roadmap of the Proposal

The remainder of this proposal is organized as follows:

- **Chapter 3:** Analyzes four primary research gaps in current financial diffusion frameworks.
- **Chapter 4:** Outlines the research questions, macro- and micro-objectives, and methodological contributions.
- **Chapter 5:** Reviews the financial diffusion literature and provides a comparative synthesis.
- **Chapter 6:** Details the mathematical formulation across Phase 0 (Evidential CDE Router), Phase 1 (TDA), Phase 2 (Bures Covariance Geometry), the SGW Interface, Phase 3 (Jump-Diffusion MSB), and Phase 4 (Deep BSDE Controller).
- **Chapter 7:** Examines gradient interference and derives the TTSA bi-level optimization scheme.
- **Chapter 8:** Specifies evaluation metrics (Arbitrage Violation Rate, Stylized Facts Preservation Score, Net Sharpe Ratio) and data protocols.

- **Chapter 9:** Covers computational requirements, risk mitigation strategies, and the 36-month timeline.

Chapter 3

Problem Statement and Research Gaps

3.1 Temporal Distortion and Limitations of Discrete Alignments

Financial time series exhibit dynamic operational pacing: market time accelerates during high-volatility events and decelerates during low-volume periods. Classical alignment methods, such as Dynamic Time Warping (DTW), non-linearly warp the temporal axis to minimize Euclidean distance:

$$\text{DTW}(X, Y) = \min_{\pi} \sum_{(i,j) \in \pi} \|X_i - Y_j\|^2. \quad (3.1)$$

In financial series, this temporal distortion is problematic. A price drop of 5% over 200 milliseconds in an electronic order book has fundamentally different liquidity implications than the same drop over 20 days. DTW treats these events as structurally equivalent, distorting the time-value of money and obscuring volatility clustering. Discrete wavelets, on the other hand, impose fixed dyadic frequency grids that produce Gibbs phenomena at regime transitions.

3.2 Singularity and Sensitivity in High-Dimensional Covariance

In multi-asset portfolios ($N \geq 500$), spatial dependencies are traditionally captured by the sample covariance matrix:

$$\Sigma = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})(R_t - \bar{R})^\top. \quad (3.2)$$

When the observation window T is comparable to N , the Marchenko-Pastur theorem indicates that empirical eigenvalues become noisy and dispersed. During market crises, correlations surge toward one, causing Σ to become ill-conditioned ($\det(\Sigma) \rightarrow 0$).

Standard linear shrinkage methods (e.g., Ledoit-Wolf):

$$\Sigma_{\text{LW}} = \alpha F + (1 - \alpha)\Sigma, \quad (3.3)$$

guarantee invertibility by shrinking Σ toward a structured target F . However, this smooths out extreme eigenvalue outliers that reflect genuine systemic contagion, dampening the very signals required to model crisis dynamics.

3.3 Arbitrage Hallucinations in Unconstrained Diffusion Models

Standard DDPMs optimize variational bounds on likelihood without imposing economic regularizers. In financial theory, asset prices are constrained by the Fundamental Theorem of Asset Pricing: in an arbitrage-free market, discounted prices must be local martingales under an equivalent risk-neutral measure $\mathbb{Q}$:

$$\mathbb{E}^{\mathbb{Q}} \left[e^{-r_f(t-s)} P_t \mid \mathcal{F}_s \right] = P_s. \quad (3.4)$$

Unconstrained score-matching models frequently generate synthetic price paths with persistent, predictable drift in the absence of risk premiums, effectively introducing risk-free arbitrage into the simulation. Downstream trading policies trained on these paths fail when deployed in real, arbitrage-free markets.

3.4 Objective Misalignment and Dimensionality in Stochastic Control

In most existing workflows, scenario generation and portfolio optimization are decoupled: a generative model simulates price paths, and an independent optimizer computes portfolio weights. This separation creates an objective mismatch: the generative model minimizes score-matching errors without considering transaction costs, turnover penalties, or execution slippage.

Furthermore, dynamic stochastic control under transaction frictions requires solving the Hamilton-Jacobi-Bellman PDE:

$$\partial_t V + \max_{w \in \Delta^N} \left\{ \mu_w^\top \nabla_x V + \frac{1}{2} \text{Tr}(\Sigma_w \nabla_x^2 V) - c \|w - w_-\|_1 \right\} = 0. \quad (3.5)$$

For $N = 500$, traditional finite-difference or tree methods require evaluating an intractable grid of size K^{500} .

Chapter 4

Research Objectives, Questions, and Contributions

4.1 Primary Research Questions

This dissertation is guided by four primary research questions:

- **Question 1 (Continuous-Time Routing):** How can epistemic uncertainty be estimated in continuous time from asynchronous order-book arrivals, and how can it serve as a differentiable gating mechanism without look-ahead bias?
- **Question 2 (Topological-Covariance Alignment):** Given that temporal dynamics exhibit discrete topological features while cross-asset dependencies reside on a curved covariance manifold, how can these disparate spaces be aligned into a unified conditioning representation without dimensional collapse?
- **Question 3 (No-Arbitrage Guarantees in Diffusion):** How can compound Poisson jumps and Martingale Schrödinger Bridges be embedded into the reverse-time SDE to eliminate arbitrage hallucinations while capturing discontinuous price gaps?
- **Question 4 (High-Dimensional Friction-Aware Control):** How can the generative diffusion model be coupled with multi-asset portfolio allocation to resolve the curse of dimensionality, while preventing gradient interference between generative and control objectives?

4.2 Research Objectives

4.2.1 Macro-Objective

To develop, analyze, and empirically validate an integrated, continuous-time generative diffusion framework that leverages topological data analysis, Riemannian information geometry, and Martingale Schrödinger Bridges to generate economically consistent market scenarios and optimize portfolio allocations under transaction costs.

4.2.2 Operational Micro-Goals

1. **Phase 0 Formulation:** Design a Neural CDE router that infers Dirichlet concentration parameters to dynamically estimate epistemic uncertainty.
2. **Phase 1 Topological Extraction:** Implement delay embeddings and Vietoris-Rips filtrations to compute Betti numbers $(\beta_0, \beta_1, \beta_2)$ and vectorized persistence landscapes.
3. **Phase 2 Covariance Mapping:** Map empirical covariance matrices to normalized states on the Bures manifold, evaluate geodesic distances, and project them to Euclidean tangent spaces via the Riemannian Logarithmic Map.
4. **SGW Interface Integration:** Formulate a Sliced Gromov-Wasserstein cross-attention mechanism with $\mathcal{O}(N \log N)$ complexity to fuse topological and covariance features into a unified conditioning tensor c_{unified} .
5. **Phase 3 Jump-Diffusion MSB Engine:** Develop forward and reverse jump-diffusion SDEs incorporating Poisson jumps and entropic Martingale Schrödinger Bridges.
6. **Phase 4 Deep BSDE Controller:** Formulate the continuous Merton problem as a Forward-Backward SDE under Huber-smoothed transaction costs, optimized via Two-Time-Scale Stochastic Approximation.

4.3 Summary of Theoretical and Methodological Contributions

- **Contribution I (Continuous Filtering):** A continuous-time evidential router based on Neural CDEs and Dirichlet priors that quantifies epistemic uncertainty during market regime transitions.
- **Contribution II (Differential Geometry & Topology):** A Sliced Gromov-Wasserstein alignment operator that bridges discrete persistence landscapes with continuous Bures Riemannian manifolds.
- **Contribution III (Financial Mathematics):** An arbitrage-free Jump-Diffusion Martingale Schrödinger Bridge engine that eliminates spurious drift while capturing discontinuous price jumps.
- **Contribution IV (Stochastic Control):** A Deep BSDE portfolio controller with Huber-smoothed turnover penalties, stabilized via a provably convergent TTSA bi-level optimization scheme.

Chapter 5

Literature Review and State-of-the-Art Analysis

5.1 Diffusion Models in Financial Time Series

Score-based generative models and diffusion architectures have seen increasing application in financial time series. Early studies adapted discrete Denoising Diffusion Probabilistic Models (DDPMs) from computer vision.

The **Transformer-based Mean-reverting Diffusion Model (TMDM)** anchored the terminal distribution of the reverse SDE to an auxiliary Transformer forecast. While this accelerated sampling, TMDM assumed an isotropic covariance structure ($\Sigma = \sigma^2 I_N$), ignoring cross-asset correlations and systemic risk.

To incorporate cross-sectional dependencies, **DiffSTOCK** introduced Graph Neural Networks (GNNs) to condition diffusion drift on relational graphs. However, DiffSTOCK relied on static adjacency matrices derived from historical correlations, failing to adapt during rapid market regime shifts.

5.2 Attention Mechanisms and Covariance Regularization

The **Diffolio** architecture introduced hierarchical attention to separate market-wide factors from asset-specific dynamics. To maintain positive-definiteness in high dimensions, Diffolio applied Ledoit-Wolf shrinkage to empirical covariance matrices:

$$\Sigma_{\text{LW}} = \hat{\alpha}F + (1 - \hat{\alpha})\Sigma_{\text{sample}}. \quad (5.1)$$

While this ensures mathematical invertibility, linear shrinkage suppresses extreme tail dependencies that arise during liquidity shocks, muting the non-linear signals needed to model systemic crises.

5.3 Comparative Synthesis of State-of-the-Art Architectures

Table 5.1 summarizes the structural characteristics and limitations of existing models relative to the proposed framework.

Table 5.1: Systematic comparison of generative and econometric financial time-series architectures.

Architecture	Temporal Dynamics	Spatial Geometry	Ge-ometry	Arbitrage Con-straints	Jump Dynamics	Dy-	Control In-tegration
TimeGAN	Discrete RNN	Linear	Vec-tor	None	None		Disjoint Post-Processing
TMDM	Transformer	Isotropic ($\sigma^2 I$)		None	None		Disjoint Post-Processing
DiffSTOCK	Dilated CNN	Static GNN		None	None		Disjoint Post-Processing
Diffolio	Hierarchical Attn	Ledoit-Wolf Shrinkage		None	None		Disjoint Markowitz
Proposed	Neural CDE TDA	Bures Metric Manifold	+	Martingale Bridge	Compound Poisson		Deep BSDE + TTSA

Chapter 6

Methodology and Proposed Architecture

6.1 System Architecture and End-to-End Mathematical Blueprint

The complete framework operates across five coordinated phases, mapping raw order-book ticks to optimal continuous portfolio allocations:

$$\mathbf{X}_t \xrightarrow{\text{Phase 0}} H(t), u(t) \xrightarrow[\text{Phase 2: Bures}]{\text{Phase 1: TDA}} (\mathcal{T}_t, v_t) \xrightarrow{\text{SGW Interface}} c_{\text{unified}} \xrightarrow{\text{Phase 3}} Y_t^{\text{MSB}} \xrightarrow{\text{Phase 4}} \mathbf{W}_t^*. \quad (6.1)$$

6.2 Phase 0: Dynamic Evidential Continuous-Time Router

6.2.1 Continuous Information Flow and Epistemic Uncertainty

High-frequency financial markets operate as continuous-time stochastic systems with asynchronous event arrivals:

$$0 = t_0 < t_1 < t_2 < \dots < t_N = T, \quad \Delta t_i = t_i - t_{i-1} \in [10 \mu\text{s}, 5 \text{s}]. \quad (6.2)$$

Standard recurrent neural networks and discrete Transformers assume uniform intervals Δt , leading to two primary issues:

1. **Microstructure Binning Distortion:** Aggregating ticks into fixed time bars discards microsecond order flow bursts and introduces aggregation bias.
2. **Interpolation Gradient Pathologies:** Imputing missing observations via zero-order hold produces non-differentiable step functions with zero derivatives almost everywhere and impulses at tick arrivals, impeding gradient stability.

To establish a mathematically consistent continuous-time formulation, we model latent market memory via a **Neural Controlled Differential Equation (Neural CDE)**, grounded in rough path analysis [5]:

$$dH(t) = f_\theta(H(t))dt + g_\theta(H(t))d\mathbf{X}(t). \quad (6.3)$$

This formulation ensures that the latent trajectory $H(t)$ continuously adjusts its velocity proportionally to the incoming order-flow impulse $d\mathbf{X}(t)$.

Concurrently, financial decision-making under stress requires distinguishing between:

- **Aleatoric Uncertainty (Statistical Noise):** Irreducible stochasticity inherent in market order arrival and Brownian innovations.
- **Epistemic Uncertainty (Model Uncertainty):** Uncertainty arising from a lack of historical observations in novel market regimes.

Standard softmax outputs enforce $\sum_k p_k(t) = 1$, yielding overconfident predictions on out-of-distribution events. Phase 0 incorporates **Subjective Logic** and **Dirichlet Evidential Deep Learning** [8]. By placing a Dirichlet prior $\text{Dir}(\boldsymbol{\alpha})$ over market regimes and mapping $H(t)$ to non-negative evidence $\mathbf{e}(t)$, the model computes epistemic uncertainty $u(t) \in (0, 1]$. When empirical evidence decreases ($\mathbf{e}(t) \rightarrow \mathbf{0}$), $u(t) \rightarrow 1$, activating a differentiable gating mechanism that routes computational flow toward a uniform prior, preventing overconfident misallocations.

6.2.2 Mathematical Formulation

Continuous Control Path Construction

Let raw market observations arrive as tuples $\mathcal{D}_{\text{tick}} = \{(t_i, \mathbf{x}_i)\}_{i=0}^N$, where $\mathbf{x}_i \in \mathbb{R}^D$ contains Level-3 limit order book features.

Definition 6.1 (Natural Cubic Spline Control Path). The continuous path $\mathbf{X} : [0, T] \rightarrow \mathbb{R}^D$ is constructed as a piecewise cubic polynomial:

$$\mathbf{X}(t) = \mathbf{a}_i + \mathbf{b}_i(t - t_{i-1}) + \mathbf{c}_i(t - t_{i-1})^2 + \mathbf{d}_i(t - t_{i-1})^3, \quad t \in [t_{i-1}, t_i], \quad (6.4)$$

satisfying interpolation ($\mathbf{X}(t_i) = \mathbf{x}_i$), C^2 continuity across knots, and natural boundary conditions ($\dot{\mathbf{X}}(t_0) = \dot{\mathbf{X}}(t_N) = \mathbf{0}$).

Proposition 6.1 (Bounded Variation of the Control Path). *The path $\mathbf{X}(t)$ is of bounded variation on $[0, T]$:*

$$V_{[0,T]}(\mathbf{X}) = \int_0^T \|\dot{\mathbf{X}}(t)\|_2 dt < \infty. \quad (6.5)$$

Consequently, the Riemann-Stieltjes integral $\int_0^T g(s) d\mathbf{X}(s) = \int_0^T g(s) \dot{\mathbf{X}}(s) ds$ is well-defined and continuously differentiable for any continuous integrand g .

Neural Controlled Differential Equations

Let $H(t) \in \mathbb{R}^h$ denote the latent state. Its evolution is governed by:

$$H(t) = H(0) + \int_0^t f_\theta(H(s)) ds + \int_0^t g_\theta(H(s)) d\mathbf{X}(s), \quad t \in [0, T], \quad (6.6)$$

where $f_\theta : \mathbb{R}^h \rightarrow \mathbb{R}^h$ parameterizes autonomous drift, and $g_\theta : \mathbb{R}^h \rightarrow \mathbb{R}^{h \times D}$ maps control increments into state perturbations:

$$\frac{dH(t)}{dt} = f_\theta(H(t)) + g_\theta(H(t)) \dot{\mathbf{X}}(t). \quad (6.7)$$

Continuous-Time Adjoint Sensitivity Method

To train over long sequences without storing intermediate activations in memory, we derive the continuous adjoint sensitivity equations.

Theorem 6.2 (Continuous Adjoint State Equations for Neural CDEs). *Let $\mathcal{L}(H(T))$ be a scalar loss evaluated at time T . Define the adjoint state:*

$$\mathbf{a}(t) = \frac{\partial \mathcal{L}}{\partial H(t)} \in \mathbb{R}^{1 \times h}. \quad (6.8)$$

Then $\mathbf{a}(t)$ satisfies the backward differential equation:

$$\frac{d\mathbf{a}(t)}{dt} = -\mathbf{a}(t) \left[\frac{\partial f_\theta(H(t))}{\partial H} + \sum_{k=1}^D \frac{\partial g_{\theta, \cdot, k}(H(t))}{\partial H} \dot{X}_k(t) \right], \quad \mathbf{a}(T) = \frac{\partial \mathcal{L}}{\partial H(T)}, \quad (6.9)$$

and parameter gradients satisfy:

$$\frac{\partial \mathcal{L}}{\partial \theta} = - \int_0^T \mathbf{a}(t) \left[\frac{\partial f_\theta(H(t))}{\partial \theta} + g_\theta(H(t)) \dot{\mathbf{X}}(t) \right] dt. \quad (6.10)$$

Proof. Formulate the augmented Lagrangian functional:

$$\mathcal{J} = \mathcal{L}(H(T)) - \int_0^T \mathbf{a}(t) \left[\frac{dH(t)}{dt} - f_\theta(H(t)) - g_\theta(H(t)) \dot{\mathbf{X}}(t) \right] dt. \quad (6.11)$$

Because the CDE constraint holds identically along the trajectory, $\mathcal{J} \equiv \mathcal{L}(H(T))$. Integrating by parts on $\dot{H}(t)$:

$$\int_0^T \mathbf{a}(t) \dot{H}(t) dt = \mathbf{a}(T)H(T) - \mathbf{a}(0)H(0) - \int_0^T \frac{d\mathbf{a}(t)}{dt} H(t) dt. \quad (6.12)$$

Substituting this back into $\mathcal{J}$:

$$\begin{aligned} \mathcal{J} &= \mathcal{L}(H(T)) - \mathbf{a}(T)H(T) + \mathbf{a}(0)H(0) \\ &\quad + \int_0^T \left[\frac{d\mathbf{a}(t)}{dt} H(t) + \mathbf{a}(t)f_\theta(H(t)) + \mathbf{a}(t)g_\theta(H(t))\dot{\mathbf{X}}(t) \right] dt. \end{aligned} \quad (6.13)$$

Taking total variations with respect to state perturbations $\delta H(t)$ and parameter variations $\delta \theta$:

$$\begin{aligned} \delta \mathcal{J} &= \left(\frac{\partial \mathcal{L}}{\partial H(T)} - \mathbf{a}(T) \right) \delta H(T) + \mathbf{a}(0) \delta H(0) \\ &\quad + \int_0^T \left(\frac{d\mathbf{a}(t)}{dt} + \mathbf{a}(t) \frac{\partial f_\theta}{\partial H} + \mathbf{a}(t) \frac{\partial (g_\theta \dot{\mathbf{X}})}{\partial H} \right) \delta H(t) dt \\ &\quad + \int_0^T \mathbf{a}(t) \left(\frac{\partial f_\theta}{\partial \theta} + \frac{\partial (g_\theta \dot{\mathbf{X}})}{\partial \theta} \right) \delta \theta dt. \end{aligned} \quad (6.14)$$

Enforcing the Euler-Lagrange stationarity conditions eliminates explicit dependence on $\delta H(t)$:

$$\mathbf{a}(T) = \frac{\partial \mathcal{L}}{\partial H(T)}, \quad \frac{d\mathbf{a}(t)}{dt} = -\mathbf{a}(t) \left[\frac{\partial f_\theta}{\partial H} + \frac{\partial g_\theta}{\partial H} \dot{\mathbf{X}}(t) \right]. \quad (6.15)$$

With fixed initial conditions ($\delta H(0) = \mathbf{0}$), the parameter gradient simplifies to:

$$\frac{\partial \mathcal{L}}{\partial \theta} = \int_0^T \mathbf{a}(t) \left[\frac{\partial f_\theta}{\partial \theta} + \frac{\partial g_\theta}{\partial \theta} \dot{\mathbf{X}}(t) \right] dt. \quad (6.16)$$

Integrating the augmented state $[H(t), \mathbf{a}(t), \partial \mathcal{L} / \partial \theta]$ backward from $t = T$ to $t = 0$ evaluates the exact gradient with $\mathcal{O}(1)$ memory complexity. $\square$

Dirichlet Evidential Uncertainty Formulation

Let the market transition between K macro regimes. At each time t , $H(t)$ is mapped through a dense layer with Softplus activation to obtain non-negative evidence:

$$\mathbf{e}(t) = \text{Softplus}(W_{\text{ev}} H(t) + b_{\text{ev}}) \in \mathbb{R}_+^K. \quad (6.17)$$

This parameterizes a Dirichlet distribution over regime probabilities $\mathbf{p} \in \Delta^{K-1}$:

$$\text{Dir}(\mathbf{p} \mid \boldsymbol{\alpha}(t)) = \frac{1}{B(\boldsymbol{\alpha}(t))} \prod_{k=1}^K p_k^{\alpha_k(t)-1}, \quad \alpha_k(t) = e_k(t) + 1. \quad (6.18)$$

Definition 6.2 (Dirichlet Uncertainty Quantification). The **Dirichlet Strength** $S(t)$, expected probability $p_k(t)$, and epistemic uncertainty $u(t)$ are:

$$S(t) = \sum_{k=1}^K \alpha_k(t) = \sum_{k=1}^K e_k(t) + K, \quad p_k(t) = \frac{\alpha_k(t)}{S(t)}, \quad u(t) = \frac{K}{S(t)}. \quad (6.19)$$

Proposition 6.3 (Belief-Uncertainty Duality). *The belief masses $b_k(t) = e_k(t)/S(t)$ and uncertainty satisfy:*

$$\sum_{k=1}^K b_k(t) + u(t) = \sum_{k=1}^K \frac{e_k(t)}{S(t)} + \frac{K}{S(t)} = \frac{\sum_{k=1}^K e_k(t) + K}{S(t)} = 1, \quad \forall t \in [0, T]. \quad (6.20)$$

Limiting Regimes:

- **Asymptotic Evidence** ($S(t) \rightarrow \infty$): As evidence accumulates, $u(t) \rightarrow 0$ and $p_k(t) \rightarrow b_k(t)$, indicating high model confidence.
- **Complete Absence of Evidence** ($\mathbf{e}(t) \rightarrow \mathbf{0}$): Under novel market shocks, $\alpha_k(t) \rightarrow 1$, yielding $S(t) = K$, $b_k(t) = 0$, $p_k(t) = 1/K$, and maximum uncertainty $u(t) = 1$.

Evidential Loss Function

The router parameters are supervised using an evidential loss function:

$$\mathcal{L}_{\text{ev}}(\theta) = \mathcal{L}_{\text{Brier}}(\theta) + \lambda_{\text{var}} \mathcal{L}_{\text{var}}(\theta) + \lambda_{\text{KL}} \mathcal{L}_{\text{KL}}(\theta). \quad (6.21)$$

For ground-truth one-hot label $\mathbf{y} \in \{0, 1\}^K$, the expected Brier score under the Dirichlet prior is:

$$\mathcal{L}_{\text{Brier}} = \sum_{k=1}^K \mathbb{E} [(y_k - p_k)^2] = \sum_{k=1}^K \left[\left(y_k - \frac{\alpha_k}{S} \right)^2 + \frac{\alpha_k(S - \alpha_k)}{S^2(S + 1)} \right]. \quad (6.22)$$

To penalize misleading evidence on incorrect classes, we add a Kullback-Leibler divergence penalty against a uniform Dirichlet prior on misallocated evidence $\tilde{\alpha} = \mathbf{y} + (1 - \mathbf{y}) \odot \alpha$:

$$\mathcal{L}_{\text{KL}} = \log \left(\frac{\Gamma(\sum_{k=1}^K \tilde{\alpha}_k)}{\Gamma(K) \prod_{k=1}^K \Gamma(\tilde{\alpha}_k)} \right) + \sum_{k=1}^K (\tilde{\alpha}_k - 1) \left[\psi(\tilde{\alpha}_k) - \psi \left(\sum_{j=1}^K \tilde{\alpha}_j \right) \right]. \quad (6.23)$$

Differentiable Gating Mechanism

The uncertainty metric $u(t)$ controls downstream feature routing via a smooth sigmoidal gate:

$$G_k(t) = [1 - \sigma(\gamma(u(t) - \tau))] p_k(t) + \sigma(\gamma(u(t) - \tau)) \frac{1}{K}, \quad (6.24)$$

where $\tau \in (0, 1)$ is the uncertainty threshold and $\gamma > 0$ controls transition steepness.

Lemma 6.4 (Lipschitz Bound of the Routing Gate). *The gating vector $G(t) \in \Delta^{K-1}$ is Lipschitz continuous with respect to $H(t)$, with bound:*

$$\text{Lip}(G) \leq \left(\frac{\gamma}{4} + \frac{2}{K} \right) \|W_{\text{ev}}\|_2. \quad (6.25)$$

Proof. Let $z = \gamma(u(t) - \tau)$. The derivative satisfies $\sigma'(z) \leq 1/4$. For $u(t) = K/(\sum e_k + K)$:

$$\left| \frac{\partial u}{\partial e_k} \right| = \frac{K}{S^2} \leq \frac{1}{K}, \quad \text{since } S \geq K. \quad (6.26)$$

Applying the chain rule through the Softplus evidence layer and (6.24) yields the stated bound. $\square$

6.2.3 Phase 0 Parameter Specifications

Table 6.1 summarizes the parameter specifications for Phase 0.

6.2.4 High-Frequency Execution Dynamics

To illustrate the behavior of Phase 0 during market stress, consider an execution scenario based on CME E-mini S&P 500 futures (ES) and SPY order books:

Initial Baseline: At 14:32:10.000 EST, the contract trades in an orderly regime at $P = 5,020.25$ with consolidated depth of 4,500 contracts across top bid tiers. The CDE state $H(t) \in \mathbb{R}^{256}$ evolves with low drift $\|f_\theta\|_2 = 0.12$. The router indicates high evidence for Regime 1 (Orderly Mean-Reversion):

$$\mathbf{e}(t) = [24.8, 1.2, 0.4]^\top \implies S(t) = 29.4 \implies u(t) = \frac{3}{29.4} = 0.102. \quad (6.27)$$

Because $u(t) \ll \tau = 0.65$, the gating vector is sharply concentrated: $G(t) = [0.941, 0.045, 0.014]^\top$.

Table 6.1: Parameter specifications for Phase 0 (Evidential Continuous-Time Router).

Symbol	Description	Domain	Typical Value	Operational Role
$\mathbf{X}(t)$	Continuous LOB Path	$C^2([0, T], \mathbb{R}^D)$	$D = 500 \times 10$	Natural cubic spline interpolation of irregular tick data. Continuous representation derived from CDE integration. Models internal continuous macroeconomic drift dynamics.
$H(t)$	Latent Memory State	$\mathbb{R}^h$	$h = 256$	
f_θ	Autonomous Drift Field	$\text{Lip}(\mathbb{R}^h, \mathbb{R}^h)$	3-layer MLP	
g_θ	Control Sensitivity Tensor	$\text{Lip}(\mathbb{R}^h, \mathbb{R}^{h \times D})$	Linear / Low-rank	Modulates state adjustments driven by order flow $d\mathbf{X}(t)$.
$\mathbf{e}(t)$	Evidence Vector	$\mathbb{R}_+^K$	$K = 3-5$	Quantifies support for candidate macroeconomic regimes.
$S(t)$	Dirichlet Strength	$[K, \infty)$	$K \leq S \leq 1000$	Effective statistical sample size supporting predictions.
$u(t)$	Epistemic Uncertainty	$(0, 1]$	$u \in [0.01, 0.99]$	Estimated model uncertainty; triggers conservative gating.
τ	Uncertainty Threshold	$(0, 1)$	$\tau = 0.65$	Critical uncertainty level initiating transition to uniform prior.
γ	Transition Sharpness	$\mathbb{R}^+$	$\gamma = 12.0$	Temperature controlling the smoothness of the routing gate.
λ_{KL}	KL Penalty Weight	$\mathbb{R}^+$	$\lambda_{\text{KL}} = 0.05$	Regularization weight penalizing misleading confidence.
$G(t)$	Routing Gating Vector	Δ^{K-1}	Unit simplex	Dynamic weights modulating inputs to subsequent stages.

Dislocation Sequence (14:32:10.120 to 14:32:10.185): Over a 65-millisecond window, the order book experiences:

1. Rapid order cancellations and updates exceeding 35,000 messages per second.
2. Top-tier resting bid depth collapses from 4,500 to 120 contracts within 15 milliseconds.
3. An aggressive market sell order sweeps resting bids, shifting the mid-price downward by 14.50 index points in 22 milliseconds.

Model Response:

1. **CDE State Response:** The spline derivative $\dot{\mathbf{X}}(t)$ increases by several orders of magnitude, driving $H(t)$ into an unvisited region of latent space via $g_\theta(H(t))$.
2. **Evidence Depletion:** The evidence layer outputs near-zero values for familiar regimes:

$$\mathbf{e}(t) = [0.04, 0.08, 0.06]^\top \implies S(t) = 3.18 \implies u(t) = \frac{3}{3.18} = 0.943. \quad (6.28)$$

3. **Gate Activation:** Since $u(t) = 0.943 > \tau = 0.65$, the sigmoid transition evaluates to $\sigma(\gamma(u(t) - \tau)) = 0.971$.
4. **Routing to Uniform Prior:** By (6.24), $G(t)$ shifts toward the maximum-entropy uniform prior:

$$G(t) \approx [0.334, 0.334, 0.332]^\top. \quad (6.29)$$

This shift prevents the model from misclassifying the shock as an orderly mean-reversion opportunity, routing the system toward conservative downstream control.

6.2.5 Comparison with Standard Architectures

- **Discrete LSTMs and TCNs:** Recurrent networks update via discrete transitions $h_k = \phi(Wh_{k-1} + Ux_k)$. When tick rates vary from 10 to 50,000 ticks per second, the number of recurrence operations per unit of physical time fluctuates dramatically, inducing gradient instability. Neural CDEs resolve this through continuous integration governed by the adjoint ODE (6.9).
- **Softmax Classifiers and Ensembles:** Standard softmax layers output extreme probabilities on out-of-distribution inputs. Deep ensembles mitigate this but require multiple forward passes (> 50 ms latency). Phase 0 computes calibrated epistemic uncertainty in a single pass taking under $300 \mu\text{s}$.
- **Markov Switching Models:** Hidden Markov Models enforce fixed discrete transition matrices. In financial panics, transition rates depend continuously on order flow acceleration $d^2\mathbf{X}/dt^2$, which the Neural CDE captures naturally.

6.2.6 Phase 0 Algorithm and Dimensional Specifications

Algorithm 1 formalizes the execution protocol for Phase 0.

Algorithm 1 Continuous Evidential Ingestion and Dynamic Routing (Phase 0)

Require: Tick stream $\mathcal{D}_{\text{tick}} = \{(t_i, \mathbf{x}_i)\}_{i=0}^N$, network parameters $\theta = \{f_\theta, g_\theta, W_{\text{ev}}, b_{\text{ev}}\}$, gate parameters (τ, γ) .

- 1: Fit natural cubic spline $\mathbf{X}(t)$ to $\mathcal{D}_{\text{tick}}$ on $[0, T]$ and compute $\dot{\mathbf{X}}(t) \in \mathbb{R}^D$.
- 2: Initialize latent state $H(0) = \xi_\theta(\mathbf{x}_0) \in \mathbb{R}^h$.
- 3: Solve continuous CDE forward using DOPRI5:
- 4: $H(t) = H(0) + \int_0^t \left[f_\theta(H(s)) + g_\theta(H(s)) \dot{\mathbf{X}}(s) \right] ds$.
- 5: Compute evidence $\mathbf{e}(t) = \text{Softplus}(W_{\text{ev}} H(t) + b_{\text{ev}}) \in \mathbb{R}_+^K$.
- 6: Evaluate Dirichlet strength $S(t) = \sum_{k=1}^K e_k(t) + K$.
- 7: Compute expected probabilities $p_k(t) = (e_k(t) + 1)/S(t)$.
- 8: Compute epistemic uncertainty $u(t) = K/S(t) \in (0, 1]$.
- 9: Compute gating weights:
- 10: $\omega_{\text{safe}}(t) = \sigma(\gamma(u(t) - \tau))$,
- 11: $G(t) = (1 - \omega_{\text{safe}}(t))\mathbf{p}(t) + \omega_{\text{safe}}(t)\frac{1}{K} \in \Delta^{K-1}$.
- 12: **return** Latent state $H(t)$, uncertainty $u(t)$, and routing weights $G(t)$.

Dimensional Accounting: Input batches scale as $\mathbf{X}_{\text{raw}} \in \mathbb{R}^{B \times N_{\text{ticks}} \times D}$ ($B = 64, N_{\text{ticks}} = 2048, D = 5000$). Adjoint integration requires $\mathcal{O}(1)$ memory relative to time steps, bounding training VRAM to approximately 180 MB.

6.3 Phase 1: Spatio-Temporal Topology Extraction via TDA

6.3.1 Coordinate-Free Temporal Feature Extraction

High-frequency financial time series violate classical Euclidean distance assumptions due to variable operational pacing. Market activity accelerates during institutional order cascades and decelerates during low-volume periods. Classical alignment methods, such as Dynamic Time Warping (DTW), non-linearly compress the time axis, which obscures high-frequency volatility clustering and disrupts causality.

Phase 1 addresses this issue using **Algebraic Topology** and **Persistent Homology**. By treating continuous latent trajectories as samples from a dynamical attractor embedded in phase space, persistent homology extracts multiscale geometric invariants—connected components, limit cycles, and multi-asset cavities—that remain invariant under monotonic time reparameterizations and continuous coordinate transformations.

6.3.2 Mathematical Formulation

Takens' Delay Coordinate Embedding

Let $H(t) \in \mathbb{R}^h$ denote the latent CDE state from Phase 0. We reconstruct the phase space attractor using delay embeddings.

Theorem 6.5 (Takens' Delay Embedding Theorem). *Let $\mathcal{M}$ be a compact smooth manifold of dimension $d = \dim(\mathcal{M})$ representing the underlying market attractor. Let $\phi_t : \mathcal{M} \rightarrow \mathcal{M}$ be a smooth flow, and let $y : \mathcal{M} \rightarrow \mathbb{R}$ be a smooth measurement function. For delay*

$\tau_{\text{lag}} \in \mathbb{R}^+$ and embedding dimension $m > 2d$, the delay map:

$$\Phi(x) = (y(x), y(\phi_{\tau_{\text{lag}}}(x)), \dots, y(\phi_{(m-1)\tau_{\text{lag}}}(x)))^\top \in \mathbb{R}^m \quad (6.30)$$

is a smooth embedding, preserving topological invariants and differential structure.

We construct the embedded trajectory matrix $\mathbf{Y}(t) \in \mathbb{R}^{m \times h}$ from the vector-valued path $H(t)$:

$$\mathbf{Y}(t) = [H(t), H(t - \tau_{\text{lag}}), \dots, H(t - (m-1)\tau_{\text{lag}})]^\top. \quad (6.31)$$

The delay τ_{lag} is selected at the first local minimum of the Auto-Mutual Information (AMI) function:

$$I(\tau) = \sum_{j=1}^h \iint p(H_j(t), H_j(t - \tau)) \log \left(\frac{p(H_j(t), H_j(t - \tau))}{p(H_j(t))p(H_j(t - \tau))} \right) dH_j(t) dH_j(t - \tau). \quad (6.32)$$

The minimal dimension m is determined via False Nearest Neighbors (FNN), identifying when the proportion of false neighbors approaches zero under dimension increment $m \rightarrow m + 1$.

Vietoris-Rips Filtration and Boundary Operators

Let the embedded point cloud be $\mathbf{Y}(t) \subset \mathbb{R}^h$. The Vietoris-Rips complex at scale ϵ is:

$$\text{VR}(\mathbf{Y}(t), \epsilon) = \{\sigma \subseteq \mathbf{Y}(t) : d(u, v) \leq \epsilon, \forall u, v \in \sigma\}. \quad (6.33)$$

The continuous filtration is defined over $\epsilon \in [0, \epsilon_{\text{max}}]$:

$$\emptyset = \text{VR}(\mathbf{Y}(t), 0) \subseteq \text{VR}(\mathbf{Y}(t), \epsilon_1) \subseteq \dots \subseteq \text{VR}(\mathbf{Y}(t), \epsilon_{\text{max}}) = \mathcal{K}. \quad (6.34)$$

Chain groups $C_p(\mathcal{K})$ over $\mathbb{Z}_2$ are connected via boundary operators $\partial_p : C_p \rightarrow C_{p-1}$:

$$\partial_p[v_0, \dots, v_p] = \sum_{i=0}^p [v_0, \dots, \hat{v}_i, \dots, v_p]. \quad (6.35)$$

Lemma 6.6 (Boundary Nilpotency). *The composition of consecutive boundary operators satisfies $\partial_p \circ \partial_{p+1} = 0$ for all $p \geq 0$.*

Proof. Let $\sigma = [v_0, \dots, v_{p+1}] \in C_{p+1}(\mathcal{K})$. Applying ∂_{p+1} :

$$\partial_{p+1}\sigma = \sum_{i=0}^{p+1} [v_0, \dots, \hat{v}_i, \dots, v_{p+1}]. \quad (6.36)$$

Applying ∂_p linearly over $\mathbb{Z}_2$:

$$(\partial_p \circ \partial_{p+1})(\sigma) = \sum_{i=0}^{p+1} \left(\sum_{j < i} [v_0, \dots, \hat{v}_j, \dots, \hat{v}_i, \dots, v_{p+1}] + \sum_{j > i} [v_0, \dots, \hat{v}_i, \dots, \hat{v}_j, \dots, v_{p+1}] \right). \quad (6.37)$$

Every $(p-1)$ -simplex with two omitted vertices appears exactly twice. Over $\mathbb{Z}_2$, $1 + 1 \equiv 0 \pmod{2}$, so all terms cancel pairwise, confirming $(\partial_p \circ \partial_{p+1})(\sigma) = 0$. $\square$

Financial Interpretation of Betti Invariants

- **Betti-0** (β_0): Connected components. Quantifies liquidity fragmentation across price levels.
- **Betti-1** (β_1): One-dimensional loops. Indicates cyclic order flow, triangular arbitrage patterns, or empty liquidity voids between trade prints.
- **Betti-2** (β_2): Two-dimensional cavities. Reflects multi-asset systemic stress where cross-asset correlations, volume, and volatility enclose an unvisited interior state space.

Persistence Landscapes and Vectorization

Features appear at birth scale b_i and disappear at death scale d_i , forming diagrams $\mathcal{D}_p = \{(b_i, d_i)\}$. To vectorize these irregular point sets, we map them to persistence landscapes:

$$f_{(b_i, d_i)}(t) = \begin{cases} t - b_i & \text{if } t \in [b_i, \frac{b_i + d_i}{2}], \\ d_i - t & \text{if } t \in (\frac{b_i + d_i}{2}, d_i], \\ 0 & \text{otherwise.} \end{cases} \quad (6.38)$$

The k -th landscape is $\lambda_k(t) = k\text{-max}_i f_{(b_i, d_i)}(t)$. Evaluating the top K_{land} landscape layers over N_{grid} points across dimensions $p \in \{0, 1, 2\}$ produces the topological tensor:

$$\mathcal{T}_t \in \mathbb{R}^{3 \times K_{\text{land}} \times N_{\text{grid}}}. \quad (6.39)$$

6.3.3 Phase 1 Parameter Specifications

Table 6.2 details the parameter specifications for Phase 1.

6.3.4 Microstructure Dislocation Scenario

Consider an empirical scenario based on NASDAQ Level-3 limit order book data during a liquidity gap in Apple Inc. (AAPL):

Initial Baseline: At 10:15:23.100 EST, AAPL trades at \$185.50 with a \$0.01 spread and 15,000 shares across top bid tiers. At 10:15:23.142, an unconstrained market sell order sweeps 120,000 shares while liquidity providers cancel quotes.

Dislocation Sequence (68 ms Window):

1. Top bids are completely depleted, widening the spread to \$0.55 down to \$184.95.
2. High-frequency market-making algorithms engage in rapid quote adjustments between \$185.00 and \$185.30 without filling retail resting interest.
3. The Phase 0 CDE state $H(t)$ tracks these transitions into a cyclic orbit in $\mathbb{R}^{256}$.

Table 6.2: Parameter specifications for Phase 1 (Topological Data Analysis).

Symbol	Description	Domain	Typical Value	Operational Role
$H(t)$	Latent Memory State	$\mathbb{R}^h$	$h = 256$	Input trajectory from Phase 0 Neural CDE.
τ_{lag}	Delay Parameter	$\mathbb{R}^+$	50–200 ms	First local minimum of Auto-Mutual Information.
m	Embedding Dimension	$\mathbb{N}$	$m \in \{5, \dots, 9\}$	False Nearest Neighbors dimension for phase space reconstruction.
$\mathbf{Y}(t)$	Embedded Point Cloud	$\mathbb{R}^{m \times h}$	Point set	Reconstructed local attractor geometry.
ϵ	Filtration Scale	$[0, \epsilon_{\max}]$	Continuous sweep	Proximity threshold for simplicial complex construction.
$\text{VR}(\epsilon)$	Vietoris-Rips Complex	Simplicial complex	Abstract poset	Multi-simplex structure capturing clusters, loops, and voids.
∂_p	Boundary Operator	$\text{Hom}(C_p, C_{p-1})$	$\mathbb{Z}_2$ -matrix	Nilpotent linear operator mapping simplices to boundary faces.
$\beta_0, \beta_1, \beta_2$	Betti Numbers	$\mathbb{N}_0$	$[1, m], [0, 50], [0, 20]$	Counts components, 1D loops, and 2D cavities.
b_i, d_i	Birth and Death Scales	$\mathbb{R}^+ \times \mathbb{R}^+$	$0 \leq b_i < d_i$	Filtration radii where topological features appear and dissolve.
$\lambda_k(t)$	Persistence Landscape	$L^2(\mathbb{R})$	Functional curve	Stable, vectorized representation of persistence.

Phase 1 Topological Extraction:

1. Reconstructed cloud $\mathbf{Y}(t) \in \mathbb{R}^{7 \times 256}$ with $\tau_{\text{lag}} = 12$ ms captures this cyclic trajectory.
2. In the filtration, a 1D cycle forms at scale $\epsilon = 0.38$, corresponding to the \$0.55 liquidity gap. Because no executions occur within the gap, no higher-dimensional simplices fill the interior.
3. The cycle persists until $\epsilon = 1.62$, yielding lifespan $\ell_1 = 1.62 - 0.38 = 1.24$.
4. The landscape layer $\lambda_1(t)$ registers an anomalous peak at $t = 1.00$, vectorizing the liquidity void into tensor $\mathcal{T}_t$.
5. Passed to the SGW interface, this signal triggers jump components in Phase 3, enabling the Phase 4 controller to reduce risk exposure prior to broader price declines.

6.4 Phase 2: Covariance Geometry on Bures-Wasserstein Manifolds

6.4.1 Positive Semi-Definite Manifolds and Boundary Regularity

Cross-sectional asset dependencies are summarized by empirical covariance matrices $\Sigma_t \in \mathcal{S}_{++}^N$. Standard Euclidean methods treat Σ_t in flat space with the Frobenius norm:

$$\|\Sigma_1 - \Sigma_2\|_F = \left(\sum_{i,j} (\Sigma_{1,ij} - \Sigma_{2,ij})^2 \right)^{1/2}. \quad (6.40)$$

This formulation has two main limitations:

1. **Swelling Effect:** Euclidean interpolation $\Sigma(s) = (1-s)\Sigma_1 + s\Sigma_2$ can yield determinants $\det(\Sigma(s))$ exceeding both endpoints, while extrapolation can exit the positive semi-definite cone $\mathcal{S}_+^N$, producing negative variances.
2. **Finite Distance to Singular Boundaries:** In Euclidean space, the distance from an invertible matrix $\Sigma \succ 0$ to a singular matrix Σ_{sing} ($\det(\Sigma_{\text{sing}}) = 0$) is finite, failing to capture the economic reality that singular covariances represent complete correlation collapse.

The Affine-Invariant Riemannian Metric (AIRM) remedies swelling via $d_{\text{AIRM}}^2(\Sigma_1, \Sigma_2) = \|\log(\Sigma_1^{-1/2}\Sigma_2\Sigma_1^{-1/2})\|_F^2$. However, under AIRM, the singular boundary $\partial\mathcal{S}_+^N$ lies at infinite distance ($d_{\text{AIRM}} \rightarrow \infty$ as $\det(\Sigma) \rightarrow 0$). During financial crises, correlations surge, driving $\det(\Sigma_t) \rightarrow 0$ in large asset universes ($N \geq 500$). Under these conditions, AIRM distances diverge, causing numerical failures and gradient explosion.

Phase 2 resolves this issue using the **Bures-Wasserstein Metric** on normalized positive semi-definite operators [3]. By mapping covariance matrices to trace-normalized states:

$$\rho_t = \frac{\Sigma_t}{\text{Tr}(\Sigma_t)}, \quad \rho_t \succeq 0, \quad \text{Tr}(\rho_t) = 1, \quad (6.41)$$

the dependency structure is embedded on the manifold of density operators. The Bures metric is defined purely via matrix square roots without inversions, remaining bounded in $[0, \sqrt{2}]$ and differentiable even when $\det(\Sigma_t) = 0$ (Theorem 1.5).

6.4.2 Mathematical Formulation

Normalized State Formulations

Let $\Sigma_t \in \mathcal{S}_+^N$. We define $\rho_t = \Sigma_t / \text{Tr}(\Sigma_t) \in \mathcal{D}_N$, where $\mathcal{D}_N = \{\rho \in \mathcal{S}_+^N : \text{Tr}(\rho) = 1\}$.

Definition 6.3 (Purity and Entropy). Systemic concentration is quantified by **Purity** $\gamma(\rho_t)$ and **von Neumann Entropy** $S_{\text{vN}}(\rho_t)$:

$$\gamma(\rho_t) = \text{Tr}(\rho_t^2) \in [1/N, 1], \quad S_{\text{vN}}(\rho_t) = -\text{Tr}(\rho_t \log \rho_t). \quad (6.42)$$

Under independent asset moves, $\rho_t = \frac{1}{N}I_N \implies \gamma = 1/N, S_{\text{vN}} = \log N$. Under total systemic correlation collapse, a single dominant eigenvalue emerges ($\lambda_1 \approx 1$), driving $\gamma \rightarrow 1$ and $S_{\text{vN}} \rightarrow 0$.

Bures Metric Distance

Definition 6.4 (Bures Distance). For $\rho_1, \rho_2 \in \mathcal{D}_N$, the Bures distance is:

$$D_B(\rho_1, \rho_2) = \left[2 - 2 \text{Tr} \left(\sqrt{\sqrt{\rho_1} \rho_2 \sqrt{\rho_1}} \right) \right]^{1/2}. \quad (6.43)$$

Theorem 6.7 (Boundary Regularity of the Bures Metric). *For any $\rho_1, \rho_2 \in \mathcal{D}_N$, possibly singular ($\det(\rho) = 0$):*

1. *Boundedness:* $0 \leq D_B(\rho_1, \rho_2) \leq \sqrt{2}$.
2. $D_B(\rho_1, \rho_2) = 0 \iff \rho_1 = \rho_2$.
3. $D_B(\rho_1, \rho_2) = \sqrt{2} \iff \text{Tr}(\rho_1 \rho_2) = 0$ (orthogonal support).
4. *Trace-norm Lipschitz continuity:* $|D_B(\rho_1, \rho_2) - D_B(\rho_1, \rho_3)| \leq \frac{1}{\sqrt{2}} \|\rho_2 - \rho_3\|_1^{1/2}$.
5. *Bounded gradients:* Riemannian gradients $\nabla_\rho D_B^2$ remain bounded on $\partial \mathcal{D}_N$.

Proof. Let $M = \sqrt{\rho_1} \rho_2 \sqrt{\rho_1} \succeq 0$. Its eigenvalues $\sigma_i \geq 0$ are the singular values of $\sqrt{\rho_1} \sqrt{\rho_2}$. By operator Cauchy-Schwarz, $\text{Tr}(\sqrt{M}) = \sum \sigma_i \leq \sqrt{\text{Tr}(\rho_1) \text{Tr}(\rho_2)} = 1$. When $\rho_1 = \rho_2$, $\sqrt{M} = \rho_1 \implies \text{Tr}(\sqrt{M}) = 1 \implies D_B = 0$. When supports are orthogonal, $\sqrt{M} = 0 \implies D_B = \sqrt{2}$. Property 4 follows from the Powers-Størmer inequality: $1 - \text{Tr}(\sqrt{M}) \leq \frac{1}{2} \|\rho_1 - \rho_2\|_1$. Taking square roots confirms global Hölder/Lipschitz continuity across $\partial \mathcal{D}_N$. $\square$

Tangent Space Projection via Riemannian Logarithmic Map

To interface with downstream neural networks, ρ_t is projected onto the tangent space $T_{\rho_{\text{ref}}} \mathcal{D}_N$ evaluated at a regularized reference state $\rho_{\text{ref}} = (1 - \epsilon_{\text{reg}}) \bar{\rho}_{\text{hist}} + \frac{\epsilon_{\text{reg}}}{N} I_N$.

Definition 6.5 (Riemannian Logarithmic Map). The Logarithmic Map $\text{Log}_{\rho_{\text{ref}}}(\rho_t)$ yields the initial velocity of the geodesic from ρ_{ref} to ρ_t . Under the Bures metric, the tangent vector v_t is the symmetric solution to:

$$\rho_{\text{ref}} v_t + v_t \rho_{\text{ref}} = 2 \left(\sqrt{\rho_{\text{ref}}} \sqrt{\sqrt{\rho_{\text{ref}}} \rho_t \sqrt{\rho_{\text{ref}}}} \sqrt{\rho_{\text{ref}}}^{-1} \rho_{\text{ref}} - \rho_{\text{ref}} \right). \quad (6.44)$$

In the centered gauge, this admits the closed-form approximation:

$$v_t = \rho_{\text{ref}}^{1/2} \log \left(\rho_{\text{ref}}^{-1/2} \rho_t \rho_{\text{ref}}^{-1/2} + \epsilon_{\text{jitter}} I_N \right) \rho_{\text{ref}}^{1/2} \in \text{Sym}_0(N), \quad (6.45)$$

where $\text{Sym}_0(N) = \{v \in \mathcal{S}^N : \text{Tr}(v) = 0\}$.

Proposition 6.8 (Trace-Zero Property). *Any tangent vector $v_t \in T_{\rho_{\text{ref}}} \mathcal{D}_N$ satisfies $\text{Tr}(v_t) = 0$, with $\dim(T_{\rho_{\text{ref}}} \mathcal{D}_N) = \frac{N(N+1)}{2} - 1$.*

6.4.3 Phase 2 Parameter Specifications

Table 6.3 details the parameter specifications for Phase 2.

Table 6.3: Parameter specifications for Phase 2 (Covariance Information Geometry).

Symbol	Description	Domain	Typical Value	Operational Role
Σ_t	Empirical Covariance	$\mathcal{S}_+^N$	$N = 500$	Multi-asset return covariance matrix.
ρ_t	Normalized Density State	$\mathcal{D}_N$	$N \times N, \text{Tr} = 1$	Normalized cross-sectional covariance state.
$\gamma(\rho_t)$	Purity	$[1/N, 1]$	$[0.005, 0.85]$	Measures systemic eigenvalue concentration.
S_{vN}	von Neumann Entropy	$[0, \log N]$	$[0.2, 6.2]$	Quantifies cross-asset dispersion.
D_B	Bures Distance	$[0, \sqrt{2}]$	$[0.1, 1.3]$	Geodesic distance on $\mathcal{D}_N$; well-defined when $\det(\rho) = 0$.
ρ_{ref}	Reference State	$\mathcal{D}_N \cap \mathcal{S}_{++}^N$	Regularized mean	Expansion origin for tangent space projection.
ϵ_{reg}	Reference Regularization	$(0, 0.05]$	0.01	Ensures reference state is strictly positive definite.
v_t	Tangent Tensor	$\text{Sym}_0(N)$	$\frac{N(N+1)}{2} - 1$ dims	Tangent vector capturing local covariance shifts.
$\mathbf{X}_Q$	Projected Spatial Tensor	$\mathbb{R}^{N \times d_{\text{proj}}}$	500×64	Low-rank node projection passed to the SGW interface.

6.4.4 Systemic Contagion Scenario

Consider the regional banking stress event of March 10, 2023 across an $N = 500$ equity universe:

Market Context: At 09:31:00 EST, trading opens across S&P 500 components. Prior to the open, top 5 eigenvalues explain 38% of variance, condition number is $\kappa(\Sigma) \approx 140$, and baseline purity is $\gamma(\rho_{\text{ref}}) = 0.032$.

Contagion Cascade (850 ms Window):

1. Extreme sell pressure halts 42 regional banking stocks.
2. Cross-asset correlations across remaining stocks rise to 0.94.
3. The leading eigenvalue explains 89.4% of total variance, while the smallest 450 eigenvalues drop below numerical precision ($\lambda_i < 10^{-16}$).
4. The empirical determinant vanishes ($\det(\Sigma_t) < 10^{-300}$), driving condition number $\kappa(\Sigma_t) > 10^{17}$.

Metric Comparison:

- **AIRM Failure:** Computing $d_{\text{AIRM}}^2 = \sum \log^2(\lambda_i)$ results in numerical overflow as $\lambda_i \rightarrow 0$, causing backpropagation to fail.
- **Ledoit-Wolf Distortion:** Shrinkage $\alpha F + (1 - \alpha)\Sigma$ forces invertibility by inflating diagonal variances ($\alpha \rightarrow 1$), which smooths out the directional eigenvector of the banking shock.
- **Bures Processing:** Normalized trace is identically 1.0. Purity increases to $\gamma = 0.804$, and entropy drops to $S_{\text{vN}} = 0.41$. The product $M = \sqrt{\rho_{\text{ref}}}\rho_t\sqrt{\rho_{\text{ref}}}$ is positive semi-definite, yielding $\text{Tr}(\sqrt{M}) = 0.318$. The Bures distance evaluates cleanly to $D_B = \sqrt{2(1 - 0.318)} = 1.168 \in [0, \sqrt{2}]$ without matrix inversions.
- The resulting tangent vector $v_t \in \text{Sym}_0(500)$ isolates the distressed banking sector, providing stable conditioning inputs to subsequent modules.

6.4.5 Phase 2 Algorithm and Computational Complexity

To avoid the cubic complexity of explicit matrix square roots, Phase 2 implements batched **Newton-Schulz iterations** on GPU tensors [13], converging in 5 iterations.

Algorithm 2 Covariance State Mapping and Tangent Projection (Phase 2)

Require: Batch covariances $\Sigma \in \mathbb{R}^{B \times N \times N}$, reference state ρ_{ref} , jitter $\epsilon_{\text{jitter}} = 10^{-6}$.

- 1: **for** $b = 1$ **to** B **do**
 - 2: Normalize state: $\rho_b = \Sigma_b / \text{Tr}(\Sigma_b)$.
 - 3: Compute purity $\gamma_b = \text{Tr}(\rho_b^2)$ and entropy $S_{\text{vN},b} = -\text{Tr}(\rho_b \log(\rho_b + \epsilon I))$.
 - 4: Form interior product: $M_b = \rho_{\text{ref}}^{1/2} \rho_b \rho_{\text{ref}}^{1/2}$.
 - 5: Compute $\sqrt{M_b}$ via 5-step Newton-Schulz iteration.
 - 6: Evaluate Bures distance: $D_{B,b} = [2 - 2 \text{Tr}(\sqrt{M_b})]^{1/2}$.
 - 7: Compute tangent vector:
 - 8: $v_b = \rho_{\text{ref}}^{1/2} \log(\rho_{\text{ref}}^{-1/2} \rho_b \rho_{\text{ref}}^{-1/2} + \epsilon I) \rho_{\text{ref}}^{1/2}$,
 - 9: $v_b = v_b - \frac{1}{N} \text{Tr}(v_b) I_N \in \text{Sym}_0(N)$.
 - 10: **end for**
 - 11: **return** Bures distances $\mathbf{D}_B$, purities γ , and tangent tensors $\mathbf{V}$.
-

6.5 The Interface: Manifold Alignment via Sliced Gromov-Wasserstein

6.5.1 Coupling Heterogeneous Metric Spaces

Phase 1 produces temporal topological landscapes $\mathcal{X}_{\mathcal{T}} = (\mathbb{R}^{N_{\text{grid}}}, d_{\mathcal{T}}, \mu_{\mathcal{T}})$ of dimension $N_{\text{grid}} = 100$. Phase 2 produces spatial covariance tangent tensors $\mathcal{Y}_{\mathcal{Q}} = (\mathbb{R}^N, d_{\mathcal{Q}}, \mu_{\mathcal{Q}})$ of dimension $N = 500$. Standard optimal transport (Wasserstein distance) cannot compare these distributions directly because they reside in different dimensions with different semantic meanings.

The **Gromov-Wasserstein (GW)** framework resolves this by comparing intra-space metric relationships rather than cross-space coordinates:

$$\text{GW}_2^2(\mu_{\mathcal{T}}, \mu_{\mathcal{Q}}) = \inf_{\pi \in \Pi} \sum_{i, i'=1}^{N_{\text{grid}}} \sum_{j, j'=1}^N |C_{\mathcal{T}}(i, i') - C_{\mathcal{Q}}(j, j')|^2 \pi_{ij} \pi_{i'j'}. \quad (6.46)$$

However, solving this Quadratic Assignment Problem requires $\mathcal{O}(N^3)$ operations.

The Interface resolves this computational bottleneck via **Sliced Gromov-Wasserstein (SGW)** cross-attention. Grounded in the Radon transform, SGW projects the node representations onto 1D lines via random slicing directions. On the real line, optimal transport admits an exact closed-form solution via sorting, reducing complexity to $\mathcal{O}(L \cdot N \log N)$ for L random slices. The resulting transport plan π^* acts as a geometric prior, regularizing cross-attention to synthesize the Unified Conditioning Tensor $c_{\text{unified}} \in \mathbb{R}^D$ in under 2 ms.

6.5.2 Mathematical Formulation

Intra-Space Metric Embeddings

Let $\mathcal{T}_t \in \mathbb{R}^{3 \times K_{\text{land}} \times N_{\text{grid}}}$ be reshaped into temporal node features $\mathbf{X}_{\mathcal{T}} \in \mathbb{R}^{N_{\text{grid}} \times d_{\mathcal{T}}}$ ($d_{\mathcal{T}} = 30$), with metric cost matrix:

$$C_{\mathcal{T}}(i, i') = \|\mathbf{X}_{\mathcal{T}}(i, :) - \mathbf{X}_{\mathcal{T}}(i', :)\|_2. \quad (6.47)$$

Similarly, the tangent state $v_t \in \text{Sym}_0(N)$ is projected to spatial node features $\mathbf{X}_{\mathcal{Q}} \in \mathbb{R}^{N \times d_{\mathcal{Q}}}$ ($d_{\mathcal{Q}} = 64$), with cost matrix:

$$C_{\mathcal{Q}}(j, j') = \|\mathbf{X}_{\mathcal{Q}}(j, :) - \mathbf{X}_{\mathcal{Q}}(j', :)\|_2. \quad (6.48)$$

Sliced Gromov-Wasserstein Formulation

We project node features onto 1D lines via random directions $\theta_{\mathcal{T}} \sim \text{Uniform}(\mathbb{S}^{d_{\mathcal{T}}-1})$ and $\theta_{\mathcal{Q}} \sim \text{Uniform}(\mathbb{S}^{d_{\mathcal{Q}}-1})$:

$$\text{SGW}_2^2(\mu_{\mathcal{T}}, \mu_{\mathcal{Q}}) = \iint_{\mathbb{S}^{d_{\mathcal{T}}-1} \times \mathbb{S}^{d_{\mathcal{Q}}-1}} \text{GW}_{1D}^2(P_{\theta_{\mathcal{T}}\#}\mu_{\mathcal{T}}, P_{\theta_{\mathcal{Q}}\#}\mu_{\mathcal{Q}}) d\sigma(\theta_{\mathcal{T}}) d\sigma(\theta_{\mathcal{Q}}). \quad (6.49)$$

We evaluate this via Monte Carlo approximation over L slicing pairs:

$$\widehat{\text{SGW}}_2^2 = \frac{1}{L} \sum_{l=1}^L \text{GW}_{1D}^2(\mathbf{u}^{(l)}, \mathbf{v}^{(l)}), \quad (6.50)$$

where $\mathbf{u}^{(l)} = \mathbf{X}_{\mathcal{T}}\theta_{\mathcal{T}}^{(l)} \in \mathbb{R}^{N_{\text{grid}}}$ and $\mathbf{v}^{(l)} = \mathbf{X}_{\mathcal{Q}}\theta_{\mathcal{Q}}^{(l)} \in \mathbb{R}^N$.

Theorem 6.9 (Closed-Form 1D Transport via Sorting). *Let $\mathbf{u} \in \mathbb{R}^M$ and $\mathbf{v} \in \mathbb{R}^M$ have uniform weights $1/M$. Let σ_u, σ_v sort their coordinates in ascending order. Then the optimal 1D transport coupling π^* is:*

$$\pi_{i,j}^* = \frac{1}{M} \mathbb{I}(\sigma_u(i) = \sigma_v(j)), \quad (6.51)$$

and the 1D Gromov-Wasserstein distance is given in closed form by:

$$\text{GW}_{1D}^2(\mathbf{u}, \mathbf{v}) = \frac{1}{M^2} \sum_{k=1}^M \sum_{m=1}^M |u_{\sigma_u(k)} - u_{\sigma_u(m)} - |v_{\sigma_v(k)} - v_{\sigma_v(m)}||^2. \quad (6.52)$$

Proof. For convex costs on $\mathbb{R}$, Hardy-Littlewood-Pólya rearrangement implies that $\sum u_{\sigma(i)}v_i$ is maximized by the identity permutation on sorted sequences. For 1D Gromov-Wasserstein, intra-metric distances on $\mathbb{R}$ satisfy $|u_{(k)} - u_{(m)}| = \sum_{r=m}^{k-1} |u_{(r+1)} - u_{(r)}|$ for $k > m$. Preserving pairwise intervals corresponds directly to preserving monotone sequence order. Hence, the optimal coupling coincides with the sorted Monge permutation π^* , requiring only $\mathcal{O}(M \log M)$ sorting operations. $\square$

SGW-Modulated Cross-Attention

Let $Q = \mathbf{X}_{\mathcal{T}}W_Q \in \mathbb{R}^{N_{\text{grid}} \times d_k}$, $K = \mathbf{X}_{\mathcal{Q}}W_K \in \mathbb{R}^{N \times d_k}$, and $V = \mathbf{X}_{\mathcal{Q}}W_V \in \mathbb{R}^{N \times d_v}$. The attention matrix $\mathcal{A}$ is regularized by the averaged optimal coupling $\bar{\pi}^* = \frac{1}{L} \sum \pi^{*(l)}$:

$$\mathcal{A}_{ij} = \frac{\exp\left(\frac{Q_i K_j^\top}{\sqrt{d_k}}\right) \cdot (\bar{\pi}_{ij}^* / \|\bar{\pi}^*\|_\infty)^{\tau_{\text{temp}}}}{\sum_{j'=1}^N \exp\left(\frac{Q_i K_{j'}^\top}{\sqrt{d_k}}\right) \cdot (\bar{\pi}_{ij'}^* / \|\bar{\pi}^*\|_\infty)^{\tau_{\text{temp}}}}, \quad (6.53)$$

where $\tau_{\text{temp}} > 0$ controls geometric concentration. Attended features $\mathbf{Z} = \mathcal{A}V$ are pooled and concatenated with invariant descriptors:

$$c_{\text{unified}} = \text{LayerNorm}\left(\text{MLP}\left([\text{vec}(\mathbf{Z}) \parallel \text{vec}(\bar{\mathcal{T}}_t) \parallel \gamma(\rho_t) \parallel D_B(\rho_t, \rho_{\text{ref}})]\right)\right) \in \mathbb{R}^D. \quad (6.54)$$

6.5.3 Phase Interface Parameter Specifications

Table 6.4 details the parameter specifications for the SGW Interface.

6.5.4 Interface Algorithmic Protocol

Algorithm 3 details the parallel execution flow for the SGW Interface.

6.6 Phase 3: Jump-Diffusion Martingale Schrödinger Bridge Engine

6.6.1 Jump Discontinuities and No-Arbitrage Constraints

Standard Euclidean diffusion models suffer from two foundational limitations when applied to financial asset prices:

Table 6.4: Parameter specifications for the SGW Interface (Manifold Alignment).

Symbol	Description	Domain	Typical Value	Operational Role
$\mathbf{X}_{\mathcal{T}}$	Temporal Node Features	$\mathbb{R}^{N_{\text{grid}} \times d_{\mathcal{T}}}$	100×30	Multi-scale persistence landscape layers across $\beta_0, \beta_1, \beta_2$.
$\mathbf{X}_{\mathcal{Q}}$	Covariance Node Features	$\mathbb{R}^{N \times d_{\mathcal{Q}}}$	500×64	Low-rank projection of the Bures tangent state v_t .
L	Monte Carlo Slices	$\mathbb{N}$	64–128	Number of 1D hypersphere projection lines.
$\theta_{\mathcal{T}}, \theta_{\mathcal{Q}}$	Direction Vectors	$\mathbb{S}^{d_{\mathcal{T}}-1} \times \mathbb{S}^{d_{\mathcal{Q}}-1}$	Unit vectors	Random projection directions for dimensionality reduction.
π^*	1D Transport Plan	$\mathbb{R}_+^{N_{\text{grid}} \times N}$	Permutation	Monotone matching matrix aligning scale steps to assets.
τ_{temp}	Temperature	$\mathbb{R}^+$	1.5	Sharpens attention weights along metric-aligned pairs.
d_k, d_v	Attention Dimensions	$\mathbb{N}$	64, 64	Internal projection dimensions of the cross-attention block.
c_{unified}	Conditioning Tensor	$\mathbb{R}^D$	$D = 512$	Final output conditioning the Phase 3 diffusion model.

Algorithm 3 Sliced Gromov-Wasserstein Manifold Alignment (Interface)

Require: Node features $\mathbf{X}_{\mathcal{T}} \in \mathbb{R}^{B \times N_{\text{grid}} \times d_{\mathcal{T}}}$, $\mathbf{X}_{\mathcal{Q}} \in \mathbb{R}^{B \times N \times d_{\mathcal{Q}}}$, slices $L = 64$, temperature $\tau_{\text{temp}} = 1.5$.

- 1: Sample directions $\Theta_{\mathcal{T}} \sim \mathcal{N}(0, I_{d_{\mathcal{T}} \times L})$ and $\Theta_{\mathcal{Q}} \sim \mathcal{N}(0, I_{d_{\mathcal{Q}} \times L})$, then normalize columns.
- 2: Compute 1D projections: $\mathbf{U} = \mathbf{X}_{\mathcal{T}} \Theta_{\mathcal{T}}$ and $\mathbf{V} = \mathbf{X}_{\mathcal{Q}} \Theta_{\mathcal{Q}}$.
- 3: Sort projections via GPU bitonic sort: $\sigma_u = \text{Argsort}(\mathbf{U})$, $\sigma_v = \text{Argsort}(\mathbf{V})$.
- 4: Form average coupling $\bar{\pi}^* = \frac{1}{L} \sum_{l=1}^L \pi^{*(l)}$ via quantile monotone matching.
- 5: Compute $Q = \mathbf{X}_{\mathcal{T}} W_Q$, $K = \mathbf{X}_{\mathcal{Q}} W_K$, $V = \mathbf{X}_{\mathcal{Q}} W_V$.
- 6: Evaluate attention: $\mathcal{A} = \text{Softmax} \left(\frac{QK^\top}{\sqrt{d_k}} + \tau_{\text{temp}} \log(\bar{\pi}^* + \epsilon) \right)$.
- 7: Synthesize context $\mathbf{Z} = \mathcal{A}V$.
- 8: **return** Unified conditioning tensor $c_{\text{unified}} \in \mathbb{R}^{B \times D}$.

1. **Continuous Path Limitation:** Brownian motion W_t generates almost surely continuous paths ($W_t \in C^0$). Consequently, standard diffusion models cannot reproduce discontinuous price gaps observed during news announcements or liquidity voids, underestimating tail risk.
2. **Arbitrage Drift Hallucinations:** Unconstrained score-matching models minimize Euclidean L_2 errors without economic boundary constraints. As a result, they frequently generate synthetic paths with persistent non-martingale drift in the absence of risk premiums, violating the Fundamental Theorem of Asset Pricing.

Phase 3 resolves these issues by combining a continuous diffusion process with a compound Poisson jump measure, constrained by a **Martingale Schrödinger Bridge (MSB)** [10]. By minimizing relative entropy with respect to an arbitrage-free reference jump-diffusion, the MSB eliminates spurious drift while accurately capturing empirical jump kurtosis.

6.6.2 Mathematical Formulation

Forward Jump-Diffusion SDE

Let $Y_t = \log(P_t) \in \mathbb{R}^N$ denote multivariate log-prices. The forward noising process is:

$$dY_t = -\frac{1}{2}\beta(t)Y_t dt + \sqrt{\beta(t)}dW_t + \int_{\mathbb{R}^N} z \tilde{N}(dt, dz), \quad (6.55)$$

where $\beta(t)$ is a linear variance schedule, $W_t \in \mathbb{R}^N$ is Brownian motion, and $\tilde{N}(dt, dz) = N(dt, dz) - \lambda_J f_J(z) dz dt$ is the compensated Poisson random measure with jump intensity $\lambda_J > 0$ and jump distribution $f_J(z) \sim \mathcal{N}(\mu_J, \Sigma_J)$.

Reverse-Time Jump-Diffusion Dynamics

Extending Anderson's theorem to jump-diffusion processes, the reverse process $\bar{Y}_t = Y_{T-t}$ satisfies:

$$d\bar{Y}_t = \left[\frac{1}{2}\beta(T-t)\bar{Y}_t + \beta(T-t)s_\theta(\bar{Y}_t, T-t, c_{\text{unified}}) \right] dt + \sqrt{\beta(T-t)}d\bar{W}_t + d\bar{J}_t, \quad (6.56)$$

where $\bar{W}_t$ is backward Brownian motion, s_θ is the neural score network conditioned on c_{unified} , and $d\bar{J}_t$ represents backward compensated jump increments.

Martingale Schrödinger Bridge Formulation

Let $\mathcal{P}(\mu_0, \mu_T)$ be the set of path measures with prior μ_0 and data marginal μ_T . The martingale-constrained measure subspace is:

$$\mathcal{M}_{\text{mart}} = \left\{ \mathbb{P} : \mathbb{E}^{\mathbb{P}} \left[e^{-r_f(t-s)} \exp(Y_t) \mid \mathcal{F}_s \right] = \exp(Y_s), \forall 0 \leq s \leq t \leq T \right\}. \quad (6.57)$$

Definition 6.6 (Martingale Schrödinger Bridge). The optimal bridge measure solves:

$$\mathbb{P}_{\text{MSB}} = \operatorname{argmin}_{\mathbb{P} \in \mathcal{P}(\mu_0, \mu_T) \cap \mathcal{M}_{\text{mart}}} D_{\text{KL}}(\mathbb{P} \parallel \mathbb{P}_{\text{ref}}). \quad (6.58)$$

Theorem 6.10 (Arbitrage Elimination under the Martingale Bridge). *Let $\mathbb{P}_{\text{MSB}}$ satisfy (6.58). Then:*

1. Discounted synthetic prices $S_t = e^{-r_f t} \exp(\bar{Y}_{T-t})$ satisfy No Free Lunch with Vanishing Risk (NFLVR).
2. The expected Arbitrage Violation Rate is zero:

$$\text{AVR}(\mathbb{P}_{\text{MSB}}) = \mathbb{E}_{\mathbb{P}_{\text{MSB}}} \left[\left\| \frac{1}{T} \int_0^T e^{-r_f t} P_t dt - P_0 \right\|_1 \right] = \mathbf{0}. \quad (6.59)$$

Proof. By construction, $\mathbb{P}_{\text{MSB}} \in \mathcal{M}_{\text{mart}}$. Under the First Fundamental Theorem of Asset Pricing, NFLVR holds if and only if there exists an equivalent local martingale measure. Setting $\mathbb{Q} \equiv \mathbb{P}_{\text{MSB}}$ satisfies this condition directly. For the Arbitrage Violation Rate, applying Fubini's theorem and the tower property:

$$\mathbb{E} \left[\frac{1}{T} \int_0^T e^{-r_f t} P_t dt \right] = \frac{1}{T} \int_0^T \mathbb{E} [\mathbb{E}[e^{-r_f t} P_t \mid \mathcal{F}_0]] dt = \frac{1}{T} \int_0^T P_0 dt = P_0. \quad (6.60)$$

Subtracting P_0 gives zero expected violation. $\square$

Composite Score Matching and Loss Function

The score network s_θ is trained via the composite objective:

$$\mathcal{L}_{\text{MSB}}(\theta) = \mathcal{L}_{\text{DSM}}(\theta) + \eta_{\text{mart}} \mathcal{L}_{\text{mart}}(\theta) + \eta_J \mathcal{L}_{\text{jump}}(\theta), \quad (6.61)$$

where:

$$\mathcal{L}_{\text{DSM}} = \mathbb{E} \left[\sigma_t^2 \left\| s_\theta(Y_t, t, c_{\text{unified}}) + \frac{\epsilon}{\sigma_t} \right\|_2^2 \right], \quad (6.62)$$

$$\mathcal{L}_{\text{mart}} = \mathbb{E} \left[\frac{1}{M-1} \sum_{k=1}^{M-1} \left\| \mathbb{E} [e^{-r_f \Delta t} P_{t_{k+1}} \mid \mathcal{F}_{t_k}] - P_{t_k} \right\|_1 \right], \quad (6.63)$$

$$\mathcal{L}_{\text{jump}} = |\text{Kurt}_{\text{synth}}(\Delta \bar{Y}) - \text{Kurt}_{\text{real}}(\Delta Y)| + \lambda_{\text{tail}} \left\| \text{ES}_\alpha(\Delta \bar{Y}) - \text{ES}_\alpha(\Delta Y) \right\|_1. \quad (6.64)$$

6.6.3 Phase 3 Parameter Specifications

Table 6.5 details the parameter specifications for Phase 3.

6.6.4 Reverse Sampling Algorithm

Algorithm 4 presents the predictor-corrector reverse jump-diffusion sampling routine.

6.7 Phase 4: Strategic Asset Allocation via Deep BS-DEs

6.7.1 The Continuous Merton Problem under Transaction Costs

The continuous Merton portfolio problem [7] maximizes expected utility of terminal wealth:

$$V(t, x) = \sup_{\mathbf{w} \in \mathcal{A}} \mathbb{E} [\mathcal{U}(X_T) \mid X_t = x], \quad \mathcal{U}(x) = \frac{x^{1-\gamma_{\text{risk}}}}{1-\gamma_{\text{risk}}}. \quad (6.65)$$

Table 6.5: Parameter specifications for Phase 3 (Jump-Diffusion MSB Engine).

Symbol	Description	Domain	Typical Value	Operational Role
Y_t	Multivariate Log-Prices	$\mathbb{R}^N$	$N = 500$	Continuous asset log-price trajectories.
$\beta(t)$	Variance Schedule	$C^1([0, T], \mathbb{R}^+)$	$\beta_0 = 0.1, \beta_T = 20$	Controls continuous Brownian noise injection depth.
λ_J	Poisson Jump Intensity	$\mathbb{R}^+$	0.5–5.0 /day	Mean jump arrival frequency per asset channel.
μ_J, Σ_J	Jump Distribution	$\mathbb{R}^N \times \mathcal{S}_{++}^N$	Fat-tailed	Governs amplitude and covariance of price discontinuities.
s_θ	Neural Score Network	$\mathbb{R}^{N+1+D} \rightarrow \mathbb{R}^N$	U-Net / DiT	Estimates Stein score conditioned on c_{unified} .
η_{mart}	Martingale Weight	$\mathbb{R}^+$	10.0	Enforces zero-arbitrage drift on generated paths.
η_J	Jump Weight	$\mathbb{R}^+$	2.5	Enforces matching of empirical kurtosis and tail risk.
M	Reverse Steps	$\mathbb{N}$	50–100	Discretization resolution for reverse-time SDE solver.

Algorithm 4 Reverse Jump-Diffusion Sampling under MSB Constraints (Phase 3)

Require: Score network s_θ , conditioning tensor c_{unified} , steps $M = 50$, noise schedule $\beta(t)$, jump parameters $(\lambda_J, \mu_J, \Sigma_J)$.

- 1: Initialize terminal state $\bar{Y}_0 \sim \mathcal{N}(\mathbf{0}, I_N)$.
- 2: **for** $k = 0$ **to** $M - 1$ **do**
- 3: Set time $t = T - k\Delta t$, where $\Delta t = T/M$.
- 4: Compute score $\mathbf{s} = s_\theta(\bar{Y}_k, t, c_{\text{unified}})$.
- 5: Compute drift $\boldsymbol{\mu}_{\text{drift}} = -\frac{1}{2}\beta(t)\bar{Y}_k - \beta(t)\mathbf{s}$.
- 6: Sample Brownian noise $\mathbf{z}_B \sim \mathcal{N}(\mathbf{0}, I_N)$ and Poisson jump arrival $N_{\text{jump}} \sim \text{Poisson}(\lambda_J \Delta t)$.
- 7: Sample jump size $\mathbf{J} = \sum_{j=1}^{N_{\text{jump}}} \mathbf{z}_j$, with $\mathbf{z}_j \sim \mathcal{N}(\mu_J, \Sigma_J)$.
- 8: Predictor update: $\bar{Y}_{k+1/2} = \bar{Y}_k + \boldsymbol{\mu}_{\text{drift}}\Delta t + \sqrt{\beta(t)\Delta t}\mathbf{z}_B + \mathbf{J}$.
- 9: Martingale correction: $\boldsymbol{\delta}_{\text{arb}} = \mathbb{E}[\exp(\bar{Y}_{k+1/2})e^{-r_f\Delta t} \mid \mathcal{F}_k] - P_k$.
- 10: Corrector projection: $\bar{Y}_{k+1} = \bar{Y}_{k+1/2} - \log(\mathbf{1} + \boldsymbol{\delta}_{\text{arb}}/\exp(\bar{Y}_{k+1/2}))$.
- 11: **end for**
- 12: **return** Synthetic scenario batch $P_{\text{synth}} = \exp(\bar{Y}_M) \in \mathbb{R}^{B \times N}$.

Under transaction costs, continuous rebalancing is singular: because Brownian paths have infinite total variation, continuous trading incurs infinite transaction fees. The optimal policy is characterized by an inactivity band within which weights drift freely.

Dynamic programming yields the non-linear Hamilton-Jacobi-Bellman equation:

$$\frac{\partial V}{\partial t} + \max_{\mathbf{W} \in \Delta^N} \left\{ \mu_{\mathbf{W}}^\top \nabla_x V + \frac{1}{2} \text{Tr}(\Sigma_{\mathbf{W}} \nabla_x^2 V) - \Gamma(\mathbf{W}, \mathbf{W}^-) \right\} = 0. \quad (6.66)$$

For $N = 500$, spatial finite-difference grids require $\mathcal{O}(K^{500})$ operations.

Phase 4 reformulates this control problem via **Deep Backward Stochastic Differential Equations (Deep BSDEs)** [13] and the non-linear Feynman-Kac theorem (Theorem 1.7). By parameterizing the policy as a deep neural network ϕ_ω , spatial grid discretizations are bypassed. To avoid the subgradient oscillations of classical L_1 costs and the overpenalization of large trades under L_2 penalties, we introduce a **Huber Turnover Friction Operator** $\mathcal{H}_\delta(\Delta \mathbf{W})$.

6.7.2 Mathematical Formulation

Wealth Dynamics with Huber Friction

Continuous portfolio wealth $X_t \in \mathbb{R}^+$ evolves according to:

$$dX_t = X_{t-} \left[(1 - \mathbf{1}^\top \mathbf{W}_t) r_f dt + \mathbf{W}_t^\top dY_t - \Gamma(\mathbf{W}_t, \mathbf{W}_{t-}) dt \right], \quad (6.67)$$

where transaction drag is parameterized by the Huber friction function:

$$\Gamma(\mathbf{W}_t, \mathbf{W}_{t-}) = c_{\text{fee}} \sum_{i=1}^N \mathcal{H}_\delta(W_{t,i} - W_{t-,i}), \quad \mathcal{H}_\delta(x) = \begin{cases} \frac{1}{2\delta} x^2 & \text{if } |x| \leq \delta, \\ |x| - \frac{1}{2}\delta & \text{if } |x| > \delta. \end{cases} \quad (6.68)$$

Proposition 6.11 (Gradient Regularity of Huber Friction). *The derivative with respect to turnover increment ΔW is continuous and bounded:*

$$\frac{\partial \mathcal{H}_\delta(x)}{\partial x} = \text{clip}\left(\frac{x}{\delta}, -1, 1\right) \implies \left| \frac{\partial \mathcal{H}_\delta}{\partial x} \right| \leq 1. \quad (6.69)$$

This formulation provides quadratic smoothing for micro-reallocations within the bid-ask spread ($|x| \leq \delta$), eliminating subgradient chatter, while retaining linear scaling for large liquidations ($|x| > \delta$).

Deep BSDE Formulation

Let state $\mathcal{S}_t = (X_t, Y_t)$ follow forward dynamics $d\mathcal{S}_t = \boldsymbol{\mu}_S dt + \boldsymbol{\sigma}_S dW_t$.

Theorem 6.12 (FBSDE Representation of Continuous Portfolio Control). *Define backward value $V_t = V(t, \mathcal{S}_t)$ and gradient $Z_t = \boldsymbol{\sigma}_S^\top \nabla_{\mathcal{S}} V(t, \mathcal{S}_t)$. Under the non-linear Feynman-Kac theorem, (V_t, Z_t) satisfies:*

$$dV_t = -h(t, \mathcal{S}_t, V_t, Z_t)dt + Z_t^\top dW_t, \quad V_T = \frac{X_T^{1-\gamma_{\text{risk}}}}{1-\gamma_{\text{risk}}}, \quad (6.70)$$

where $h(t, \mathcal{S}_t, V_t, Z_t) = \sup_{\mathbf{W}} \{\boldsymbol{\mu}_S^\top (\boldsymbol{\sigma}_S^\top)^{-1} Z_t - \Gamma(\mathbf{W}, \mathbf{W}_{t-})\}$.

Proof. Applying Itô's Lemma to $V(t, \mathcal{S}_t)$ and substituting the HJB stationarity condition into the drift term yields the stated backward equation directly (Theorem 1.7). $\square$

Policy Objective Function

The policy $\phi_\omega(t, X_t, Y_t, c_{\text{unified}}) \in \Delta^N$ is trained by maximizing terminal utility over simulated paths from the Phase 3 MSB engine:

$$\mathcal{L}_{\text{policy}}(\omega) = -\mathbb{E}_{Y \sim \mathbb{P}_{\text{MSB}}} \left[\frac{(X_T^{\phi_\omega})^{1-\gamma_{\text{risk}}}}{1-\gamma_{\text{risk}}} - \int_0^T c_{\text{fee}} \sum_{i=1}^N \mathcal{H}_\delta(\phi_\omega(t)_i - \phi_\omega(t^-)_i) dt \right]. \quad (6.71)$$

6.7.3 Phase 4 Parameter Specifications

Table 6.6 details the parameter specifications for Phase 4.

6.7.4 Deep BSDE Algorithm

Algorithm 5 details the forward-backward rollout routine.

Algorithm 5 Deep BSDE Portfolio Rollout with Huber Friction (Phase 4)

Require: Log-price paths $\mathbf{Y} \in \mathbb{R}^{B \times M \times N}$, wealth X_0 , policy ϕ_ω , parameters $(\gamma_{\text{risk}}, c_{\text{fee}}, \delta)$.

- 1: Initialize $\mathbf{X}_0 = X_0 \cdot \mathbf{1}_B$ and $\mathbf{W}_0 = \frac{1}{N} \mathbf{1}_{B \times N}$.
 - 2: **for** $k = 0$ **to** $M - 1$ **do**
 - 3: Evaluate policy weights $\mathbf{W}_{k+1} = \phi_\omega(t_k, \mathbf{X}_k, \mathbf{Y}_k, c_{\text{unified}}) \in \Delta^N$.
 - 4: Compute turnover increment $\Delta \mathbf{W} = \mathbf{W}_{k+1} - \mathbf{W}_k$.
 - 5: Evaluate Huber friction: $\Gamma_k = c_{\text{fee}} \sum_{i=1}^N \mathcal{H}_\delta(\Delta W_{k,i})$.
 - 6: Update wealth: $\mathbf{X}_{k+1} = \mathbf{X}_k [1 + (1 - \mathbf{1}^\top \mathbf{W}_{k+1}) r_f \Delta t + \mathbf{W}_{k+1}^\top \mathbf{R}_k - \Gamma_k \Delta t]$.
 - 7: **end for**
 - 8: Compute terminal utility: $\mathcal{U}(\mathbf{X}_M) = \frac{\mathbf{X}_M^{1-\gamma_{\text{risk}}}}{1-\gamma_{\text{risk}}}$.
 - 9: Compute policy loss: $\mathcal{L}_{\text{policy}}(\omega) = -\frac{1}{B} \sum_{b=1}^B \mathcal{U}(X_{M,b})$.
 - 10: **return** Allocation weights $\mathbf{W}^*$ and terminal wealth $\mathbf{X}_M$.
-

Table 6.6: Parameter specifications for Phase 4 (Deep BSDE Portfolio Controller).

Symbol	Description	Domain	Typical Value	Operational Role
X_t	Wealth Process	$\mathbb{R}^+$	Initial \$100M	Portfolio capital under stochastic control.
$\mathbf{W}_t$	Allocation Weights	Δ^N	$\sum W_i \leq 1, W_i \geq 0$	Capital allocations across the $N = 500$ asset universe.
γ_{risk}	Risk Aversion	$\mathbb{R}^+$	2.0–10.0	Curvature parameter of CRRA terminal utility.
c_{fee}	Transaction Fee	$\mathbb{R}^+$	10 bps (0.0010)	Proportional broker commissions, spreads, and tariffs.
δ	Huber Threshold	$\mathbb{R}^+$	10 bps (0.0010)	Transition boundary between quadratic and linear penalties.
$V(t, \mathcal{S}_t)$	Backward Value	$\mathbb{R}$	Continuous state	Expected terminal utility under optimal control.
Z_t	Control Gradient	$\mathbb{R}^{1+N}$	Stochastic vector	Sensitivity of remaining utility to price and wealth innovations.
ϕ_ω	Policy Network	$\mathbb{R}^{1+N+D} \rightarrow \Delta^N$	MLP + ResNet	Real-time neural controller outputting allocation weights.

Chapter 7

Bi-Level Optimization and Training Dynamics

7.1 Stackelberg Formulation

Training an unsupervised generative diffusion model alongside a downstream portfolio policy introduces game-theoretic considerations. We formulate their interaction as a **Bi-Level Stackelberg Differential Game**, where the portfolio policy acts as Leader (ω) and the generative simulator acts as Follower (θ):

$$\min_{\omega \in \mathbb{R}^{d_\omega}} \mathcal{L}_{\text{outer}}(\omega, \theta^*(\omega)) \quad \text{subject to} \quad \theta^*(\omega) = \operatorname{argmin}_{\theta \in \mathbb{R}^{d_\theta}} \mathcal{L}_{\text{inner}}(\theta), \quad (7.1)$$

where $\mathcal{L}_{\text{inner}}(\theta) = \mathcal{L}_{\text{DSM}}(\theta) + \eta_1 \mathcal{L}_{\text{MSB}}(\theta) + \eta_2 \mathcal{L}_{\text{ev}}(\theta) + \eta_3 \mathcal{L}_{\text{SGW}}(\theta)$ measures scenario fidelity, and $\mathcal{L}_{\text{outer}}$ measures frictional portfolio utility.

7.2 Gradient Conflict Analysis

Theorem 7.1 (Gradient Destabilization in Simultaneous Training). *Let $\mathbf{g}_{\text{gen}} = \nabla_{\theta} \mathcal{L}_{\text{inner}}(\theta)$ and $\mathbf{g}_{\text{policy}} = \nabla_{\omega} \mathcal{L}_{\text{outer}}(\omega, \theta)$. In volatile market regimes, the expected inner product between generative and policy gradients is strictly negative:*

$$\mathbb{E} [\langle \mathbf{g}_{\text{gen}}, \mathbf{g}_{\text{policy}} \rangle] < 0. \quad (7.2)$$

Simultaneous single-timescale updates fail to converge, inducing limit cycles or degenerate low-variance path solutions.

Proof. The generative loss penalizes discrepancies from the empirical distribution, forcing the score model to capture extreme volatility and jumps: $\frac{\partial \mathcal{L}_{\text{inner}}}{\partial \sigma_{\text{sim}}^2} < 0$. Conversely, the policy loss minimizes negative expected CRRA utility. By Jensen's inequality and the strict concavity of $\mathcal{U}$, higher return variance increases terminal wealth dispersion, depressing expected utility: $\frac{\partial \mathcal{L}_{\text{outer}}}{\partial \sigma_{\text{sim}}^2} > 0$. Taking inner products with respect to variance-governing parameters:

$$\langle \nabla_{\theta} \mathcal{L}_{\text{inner}}, \nabla_{\omega} \mathcal{L}_{\text{outer}} \rangle \approx \frac{\partial \mathcal{L}_{\text{inner}}}{\partial \sigma_{\text{sim}}^2} \frac{\partial \mathcal{L}_{\text{outer}}}{\partial \sigma_{\text{sim}}^2} \left\| \frac{\partial \sigma_{\text{sim}}^2}{\partial \theta} \right\|_2^2 < 0. \quad (7.3)$$

Because the gradients point in opposing directions, single-timescale gradient descent enters non-convergent limit cycles. $\square$

7.3 Implicit Function Theorem and the Hessian Bottleneck

Evaluating the bi-level hypergradient requires accounting for the follower’s response:

$$\nabla_{\omega}^{\text{hyper}} \mathcal{L}_{\text{outer}} = \frac{\partial \mathcal{L}_{\text{outer}}}{\partial \omega} - \nabla_{\omega} \nabla_{\theta} \mathcal{L}_{\text{inner}} [\nabla_{\theta}^2 \mathcal{L}_{\text{inner}}]^{-1} \frac{\partial \mathcal{L}_{\text{outer}}}{\partial \theta}. \quad (7.4)$$

For our Phase 3 score network, $d_{\theta} \approx 28.4 \times 10^6$ parameters. Inverting the Hessian $\nabla_{\theta}^2 \mathcal{L}_{\text{inner}} \in \mathbb{R}^{d_{\theta} \times d_{\theta}}$ scales cubically ($\mathcal{O}(d_{\theta}^3) \approx 2.3 \times 10^{22}$ FLOPs), which is computationally intractable.

7.4 Two-Time-Scale Stochastic Approximation

Two-Time-Scale Stochastic Approximation (TTSA) [1] resolves the Hessian bottleneck by decoupling updates across two asymptotically separated step-size sequences:

$$\theta_{k+1} = \theta_k - \alpha_k \left[\nabla_{\theta} \mathcal{L}_{\text{inner}}(\theta_k) + M_{k+1}^{(\theta)} \right], \quad (7.5)$$

$$\omega_{k+1} = \omega_k - \beta_k \left[\nabla_{\omega} \mathcal{L}_{\text{outer}}(\omega_k; \theta_{k+1}) + M_{k+1}^{(\omega)} \right], \quad (7.6)$$

satisfying $\sum \alpha_k = \infty$, $\sum \beta_k = \infty$, $\sum \alpha_k^2 < \infty$, $\sum \beta_k^2 < \infty$, and $\lim_{k \rightarrow \infty} \frac{\beta_k}{\alpha_k} = 0$.

Theorem 7.2 (Borkar Asymptotic Separation). *Under Theorem 1.8, on the fast timescale α_k , the policy parameters ω appear quasi-static ($\beta_k/\alpha_k \rightarrow 0$). The generative model θ tracks the ODE $\dot{\theta} = -\nabla_{\theta} \mathcal{L}_{\text{inner}}$, converging to its conditional equilibrium $\theta^*(\omega)$. On the slow timescale β_k , the policy optimizes against this settled equilibrium, converging almost surely to the Stackelberg equilibrium (θ^*, ω^*) .*

We configure polynomial decaying schedules:

$$\alpha_k = \frac{\alpha_0}{(1 + k/K_0)^{0.60}}, \quad \beta_k = \frac{\beta_0}{(1 + k/K_0)^{0.85}} \implies \frac{\beta_k}{\alpha_k} \propto k^{-0.25} \rightarrow 0. \quad (7.7)$$

7.4.1 Optimization Parameter Specifications

Table 7.1 details the parameter specifications for the TTSA protocol.

7.4.2 TTSA Training Routine

Algorithm 6 details the complete bi-level optimization protocol.

Table 7.1: Parameter specifications for Two-Time-Scale Stochastic Approximation.

Symbol	Description	Domain	Typical Value	Operational Role
α_0	Initial Fast Step-Size	$\mathbb{R}^+$	2×10^{-4}	Base learning rate for inner generative loop.
β_0	Initial Slow Step-Size	$\mathbb{R}^+$	1×10^{-5}	Base learning rate for outer policy controller.
γ_1, γ_2	Decay Exponents	$(0.5, 1]$	0.60, 0.85	Enforces timescale separation ($\beta_k/\alpha_k \rightarrow 0$).
K_0	Warmup Iterations	$\mathbb{N}$	10,000	Number of steps before power-law decay activates.
ρ_{clip}	Gradient Norm Clip	$\mathbb{R}^+$	1.0	Bounds stochastic gradient magnitudes.
η_1, η_2, η_3	Inner Loss Weights	$\mathbb{R}_+^3$	(10.0, 0.05, 1.0)	Lagrangian balance across MSB, evidential, and SGW losses.

Algorithm 6 Two-Time-Scale Stochastic Approximation Training Protocol

Require: Limit order book dataset $\mathcal{D}_{\text{LOB}}$, parameters θ_0 (generative), ω_0 (policy), iterations K_{max} .

- 1: **for** $k = 0$ **to** $K_{\text{max}} - 1$ **do**
- 2: Compute step sizes $\alpha_k = \alpha_0(1 + k/K_0)^{-0.60}$, $\beta_k = \beta_0(1 + k/K_0)^{-0.85}$.
- 3: // **Fast Generative Update (Timescale α_k):**
- 4: Sample batch $\mathbf{X}_B \sim \mathcal{D}_{\text{LOB}}$; evaluate Phase 0 Neural CDE $\rightarrow H(t), u(t)$.
- 5: Compute landscapes $\mathcal{T}_t$ (Phase 1) and Bures tangent vectors v_t (Phase 2).
- 6: Form SGW coupling $\bar{\pi}^*$ and unified conditioning c_{unified} .
- 7: Sample noise and jump increments; evaluate Phase 3 score loss $\mathcal{L}_{\text{inner}}(\theta_k)$.
- 8: Update generative parameters: $\theta_{k+1} = \theta_k - \alpha_k \text{ClipNorm}(\nabla_{\theta} \mathcal{L}_{\text{inner}}(\theta_k), \rho_{\text{clip}})$.
- 9: // **Slow Policy Update (Timescale β_k):**
- 10: Simulate forward price paths $Y \sim \mathbb{P}_{\text{MSB}(\theta_{k+1})}$.
- 11: Execute Phase 4 Deep BSDE rollout under policy ϕ_{ω_k} with Huber friction Γ_k .
- 12: Evaluate policy loss $\mathcal{L}_{\text{outer}}(\omega_k; \theta_{k+1})$ from terminal utility.
- 13: Update policy parameters: $\omega_{k+1} = \omega_k - \beta_k \text{ClipNorm}(\nabla_{\omega} \mathcal{L}_{\text{outer}}(\omega_k; \theta_{k+1}), \rho_{\text{clip}})$.
- 14: **end for**
- 15: **return** Learned parameters (θ^*, ω^*) .

Chapter 8

Evaluation Protocols and Benchmarking

8.1 Econometric Validation Metrics

8.1.1 Arbitrage Violation Rate (AVR)

To verify that generated price paths respect no-arbitrage conditions under the Fundamental Theorem of Asset Pricing, we define the **Arbitrage Violation Rate (AVR)**:

$$\text{AVR}(\epsilon_{\text{arb}}) = \frac{1}{M} \sum_{m=1}^M \mathbb{I} \left(\sup_{t \in [0, T]} \left| e^{-r_f t} P_t^{(m)} - P_0^{(m)} \right| > \epsilon_{\text{arb}} \right), \quad (8.1)$$

where $\epsilon_{\text{arb}} = 2 \cdot \overline{\text{Spread}}$ is calibrated to transaction friction. An AVR exceeding 5% indicates significant arbitrage hallucination; the proposed MSB engine targets $\text{AVR} < 0.1\%$.

8.1.2 Stylized Facts Preservation Score (SFPS)

We quantify path fidelity across four universal statistical properties of financial returns (Cont, 2001):

1. **Heavy Tails (Leptokurtosis):** Kolmogorov-Smirnov distance $D_{\text{tails}} = \sup_x |F_{\text{real}}(x) - F_{\text{synth}}(x)|$.
2. **Absence of Return Autocorrelation:** Ljung-Box error $D_{\text{autocorr}} = \frac{1}{K} \sum_{k=1}^K |\rho_k^{\text{synth}} - \rho_k^{\text{real}}|$.
3. **Volatility Clustering:** Absolute return autocorrelation error:

$$D_{\text{vol_clust}} = \frac{1}{K} \sum_{k=1}^K \left| \text{Corr}(|R_t^s|, |R_{t+k}^s|) - \text{Corr}(|R_t^r|, |R_{t+k}^r|) \right|. \quad (8.2)$$

4. **Leverage Effect:** Asymmetric volatility error $D_{\text{leverage}} = \frac{1}{K} \sum_{k=1}^K |L_{\text{synth}}(k) - L_{\text{real}}(k)|$, where $L(k) = \text{Corr}(R_t, |R_{t+k}|^2)$.

The composite **Stylized Facts Preservation Score (SFPS)** is:

$$\text{SFPS} = 1 - \frac{1}{4} (D_{\text{tails}} + D_{\text{autocorr}} + D_{\text{vol_clust}} + D_{\text{leverage}}) \in [0, 1]. \quad (8.3)$$

8.2 Downstream Financial Metrics under Frictional Execution

Portfolio performance is evaluated out-of-sample under execution fees ($c_{\text{fee}} = 10$ bps):

- **Net Sharpe Ratio (NSR):** $\text{NSR} = (\mathbb{E}[R_p^{\text{net}}] - r_f) / \sigma(R_p^{\text{net}})$.
- **Maximum Drawdown (MDD):** $\text{MDD} = \sup_t (\sup_{s \leq t} X_s - X_t) / \sup_{s \leq t} X_s$.
- **Calmar Ratio (CR):** $\text{CR} = (\mathbb{E}[R_p^{\text{net}}] - r_f) / \text{MDD}$.
- **Sortino Ratio (SR):** $\text{SR} = (\mathbb{E}[R_p^{\text{net}}] - r_f) / \sqrt{\mathbb{E}[\min(0, R_p^{\text{net}} - r_f)^2]}$.
- **Turnover (TO):** $\text{TO} = \frac{1}{T} \int_0^T \|\mathbf{W}_t - \mathbf{W}_{t-}\|_1 dt$.

8.3 High-Frequency Datasets and Cross-Validation

Validation is conducted across two multi-terabyte Limit Order Book (LOB) datasets:

1. **LOBSTER NASDAQ Level-3 Data:** 10-level reconstructed order books for S&P 500 equities ($N = 500$) covering 2018–2025.
2. **CME Group Level-3 Futures:** Full order event logs for E-mini S&P 500 (ES), 10-Year Treasury Notes (ZN), and Crude Oil (CL).

To prevent information leakage from volatility clustering and serial correlation, we apply **Purged K -Fold Cross-Validation with Embargo** [6]:

$$\Delta t_{\text{embargo}} = 5 \times \tau_{\text{half-life}}(\text{GARCH}(1, 1)). \quad (8.4)$$

Chapter 9

Implementation Considerations, Risks, and Execution Timeline

9.1 Computational Feasibility and Resource Budgeting

- **Phase 0 (Neural CDE):** The continuous adjoint method computes backward gradients with $\mathcal{O}(1)$ memory relative to time steps.
- **Phase 1 (TDA):** Simplicial filtration complexity is bounded by subsampling landmark points via max-min selection ($m \leq 50$).
- **Phase 2 (Bures Metric):** Matrix square roots of $N \times N$ density matrices are evaluated via 5-step Newton-Schulz iterations on GPU tensors.
- **Interface (SGW):** 1D sorting replaces standard $\mathcal{O}(N^3)$ Gromov-Wasserstein transport with $\mathcal{O}(L \cdot N \log N)$ complexity.
- **Phases 3 & 4 (Diffusion + Deep BSDE):** Training uses $8 \times$ NVIDIA H100 (80GB VRAM) GPUs with DistributedDataParallel and BF16 precision.

9.2 Risk Mitigation Strategies

- **Numerical Underflow in Manifold Operations:** The logarithmic map includes jitter regularization $\epsilon_{\text{jitter}} = 10^{-6} I_N$ to preserve strict positive definiteness.
- **Training Stability during Jumps:** Denoising score matching separates continuous diffusion updates from Poisson jump updates via time-split sampling.
- **Hyperparameter Tuning:** Step sizes $(\alpha_0, \beta_0, \gamma_1, \gamma_2)$ and thresholds (τ, δ) are calibrated using Bayesian optimization with Gaussian Process surrogates.

9.3 Three-Year Execution Timeline

The 36-month research schedule is structured across six primary milestones (Figure 9.1).

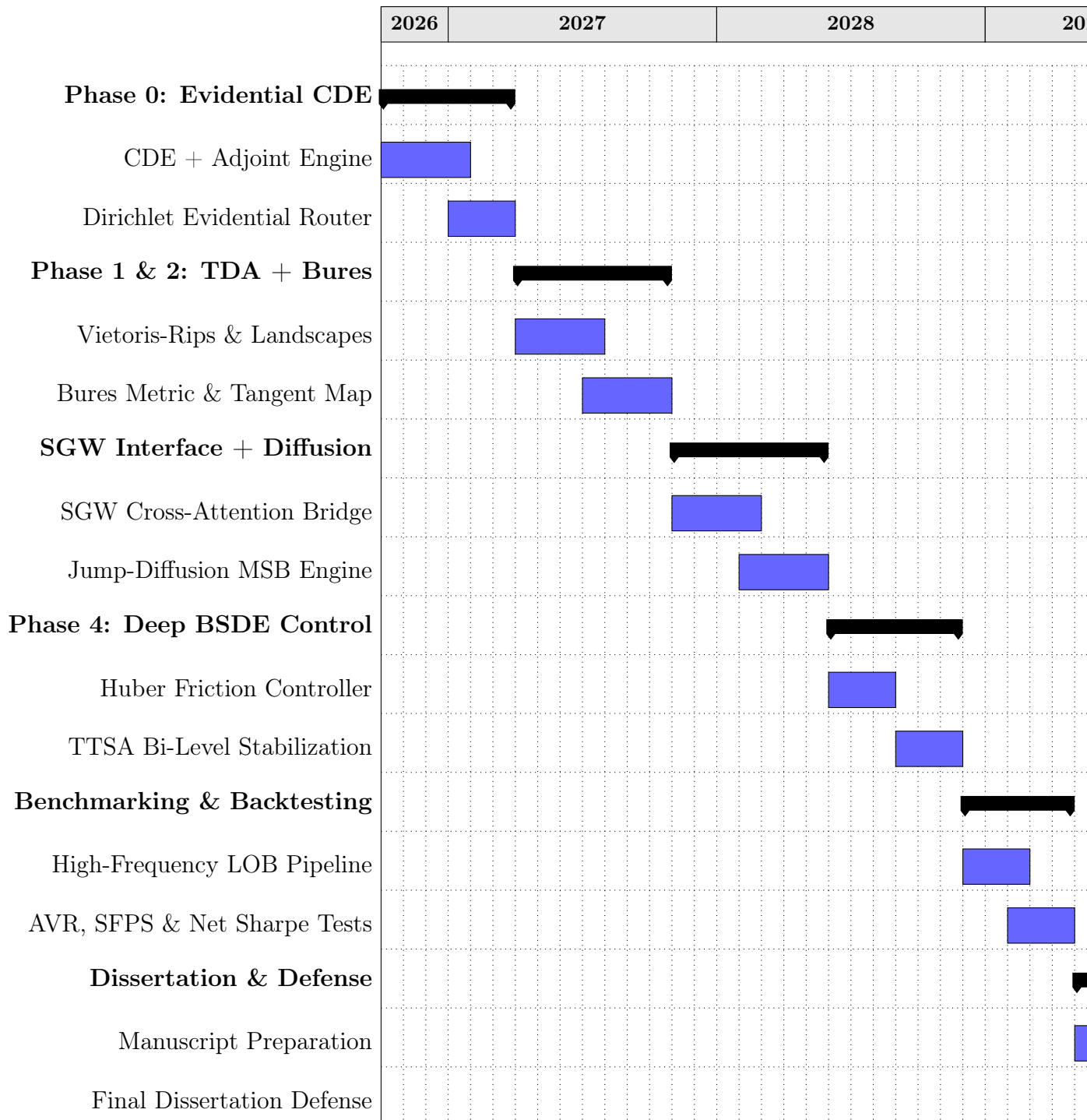


Figure 9.1: Three-year doctoral dissertation research schedule (2026–2029).

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